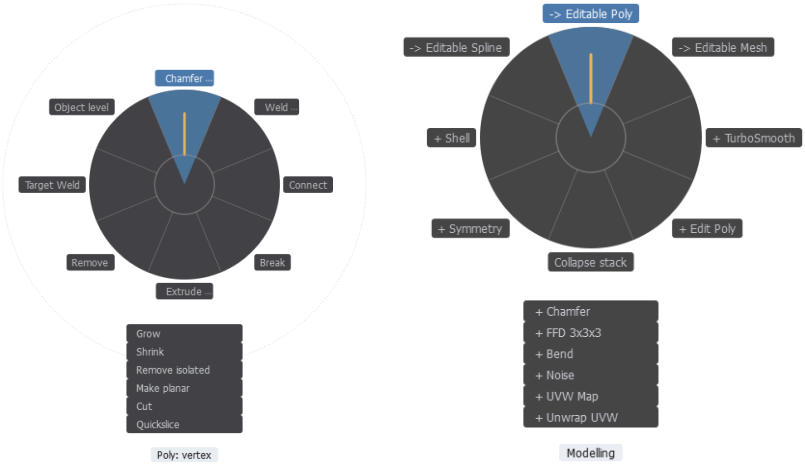
MarkingForge

QuickStart - from your first gesture to your own dials

3DS MAX 2027 · VERSION 1.5.54 · ABOUT 40 MINUTES



REAL MARKING MENUS - A QUICK GESTURE NEEDS NO DIAL

MarkingForge works the way marking menus are meant to: the command is chosen by the **direction** of the movement, not by clicking an item. Press the key and flick the mouse towards the command in one quick movement - the command runs and **the dial is not even drawn**. The dial appears only when you hold the key and wait, to remind you where things are. After a few days your hand knows the directions and you stop looking.

Twenty-two short tutorials. Part 1 teaches your hand the dials that come with MarkingForge. Part 2 shows how to change them in the menu editor: your own commands, extra rows, keys, versions for different situations, submenus, ready-made scripts, number sliders, colours, several pages on one key and the dial library. Every picture is the real program. Installing is in the **Installation Guide**; every option is explained in the **Reference**. Prefer simpler words? The **QuickStart - the easy version** teaches the same 22 lessons in plain words.

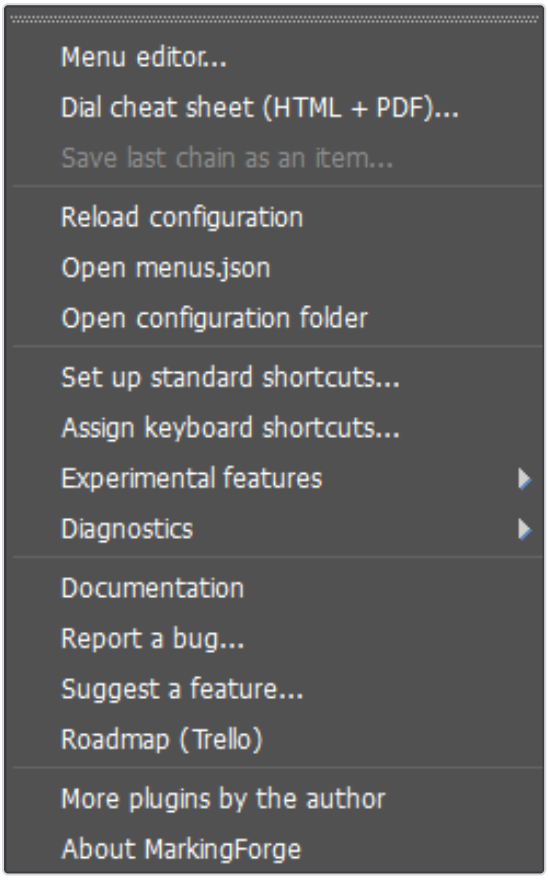
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9. Support, bug reports and feature requests

Before you begin

Check two things - it takes ten seconds:

1. There is a **MarkingForge** menu in the 3ds Max menu bar. If not, see the Installation Guide.
2. The dials have keys. The first time 3ds Max starts with MarkingForge, every dial gets its standard key - Alt, Ctrl+Alt or Shift+Alt with a digit from 1 to 8 - unless something else already uses that key. To check, open **MarkingForge > Set up standard shortcuts...**: it lists the keys it would add, or says every dial already has one.



The MarkingForge menu. Everything in this guide starts here or on a key.

Your 40 minutes

Minutes	What you do
0 - 10	Tutorials 1-6: the gestures and the built-in dials.
10 - 15	Tutorial 7: the editor's layout. Tutorial 8: your first own command on a dial.
15 - 40	Tutorials 9-22: shorter captions, extra rows, keys, versions for different situations, submenus, scripts, number sliders, colours, sharing, dials that fill themselves, pages, the dial library and the quick wizard.

Six ideas in one minute

1. A dial is a round menu under a key

Hold Alt+1 and eight commands appear around the cursor. 24 dials sit on Alt, Ctrl+Alt and Shift+Alt with the digits 1-8.

2. You aim, you do not read

Move towards a command and let go of the key. The direction is what your hand remembers - after a week you will not look.

3. Hold, flick or tap

Hold to see the dial. Flick - press and move at once - to run a command without the dial ever appearing. Release in the centre to cancel.

4. The dial follows your work

The same key shows vertex tools at vertex level, edge tools at edge level, spline tools on a shape - and material tools over the Slate editor.

5. Everything can be changed

The menu editor can put any 3ds Max command - there are over 4000 - a ready-made script, a submenu or a number slider on any direction. One key can even hold several dials, turned like pages with the mouse wheel. No restart needed.

6. It is safe to try

Cancel is always one move away, every command undoes with Ctrl+Z, and the editor changes nothing until you press Save.

PART 1

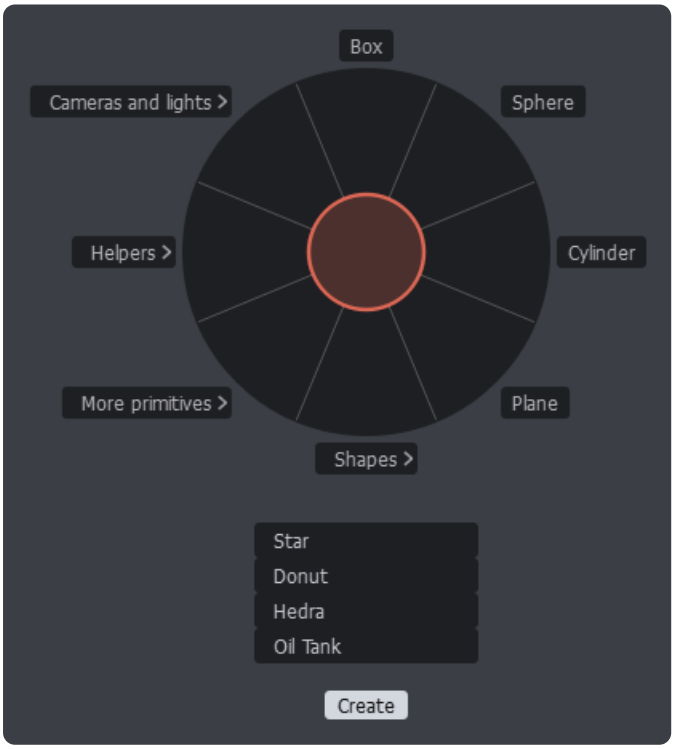
Using the dials

Six tutorials that teach your hand the dials MarkingForge comes with. Have an empty scene open.

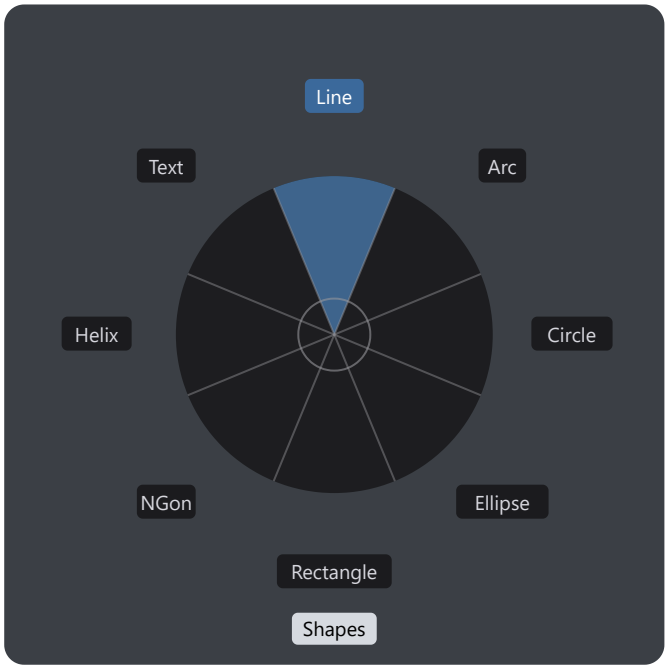
1 Your first gesture

2 min · You will learn: to hold, aim and release

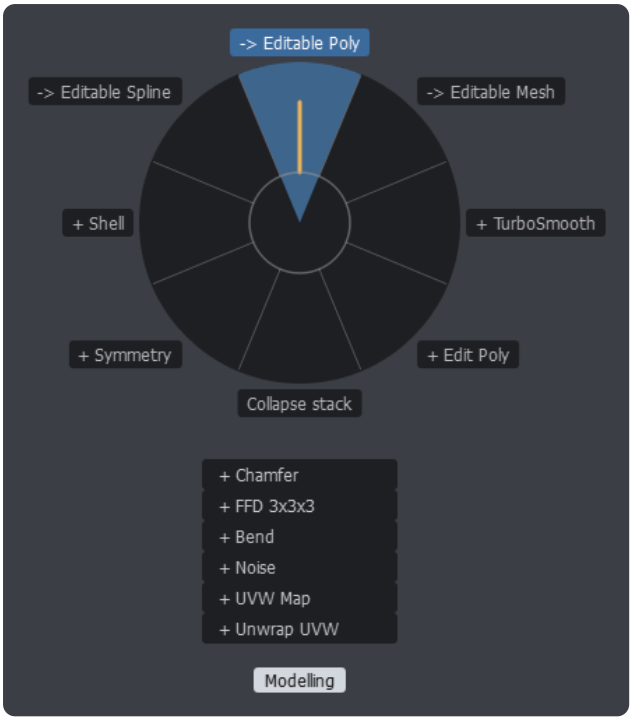
1. Move the mouse over a viewport.
2. Hold **Alt+4**. The **Create** dial appears around the cursor.
3. Move the mouse up - towards **Box**. The wedge lights up.
4. Release the keys. 3ds Max is now in Box creation: drag a box in the viewport as usual.
5. Hold **Alt+4** again and move **down**, through **Shapes >**, without letting go: the dial stays where it is and now shows the shapes - that is a **submenu**. Move up to **Line** and release. Esc finishes the line.
6. Click the box to select it, hold **Alt+1** (Modelling) over it, move up to **-> Editable Poly** and release.



Alt+4 - Create: four solids from north to south-east; four groups from south to north-west, each a submenu (marked >).



...through Shapes: the Shapes submenu, aimed at Line.



Alt+1 aimed north: the box becomes an Editable Poly.

RESULT

The box is an Editable Poly - look at the modifier stack. Two gestures, no menu, no command panel.

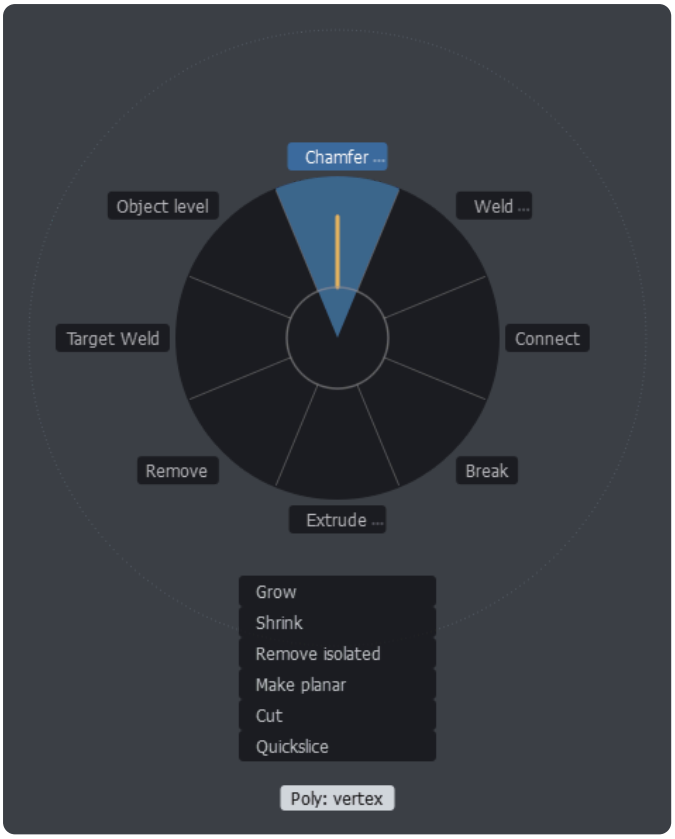
Changed your mind? Move back to the centre and release, or press Esc. Nothing runs.

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2 The dial follows you

4 min · You will learn: how one key changes with the sub-object level

1. Keep the Editable Poly selected and press **1** (vertex level).
2. Hold **Alt+1**: the same key now shows vertex tools - Chamfer, Weld, Connect, Break...
3. Aim at **Chamfer** (north) and release, then drag in the viewport.
4. Press **2** (edge level), select an edge and hold **Alt+1**: Connect, Chamfer, Loop, Ring...
5. Press **4** (polygon level), select a polygon, hold **Alt+1**, aim east - **Inset**.
6. Aim **north-west** at any level - **Object level** takes you back up.



Vertex level, aimed at Chamfer.



Polygon level, aimed at Inset.

> **One layout for both kinds of poly.** Chamfer, Extrude, Weld... sit in the same direction on an Editable Poly and under an Edit Poly modifier. Learn it once.

> **RESULT**
> You modelled at three sub-object levels with one key.

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3 Flick - the fast way

3 min · You will learn: to run commands without waiting for the dial

A marking menu is fastest when you do not wait for it. Press the key and move in the same instant, then release: the command runs, the dial is never drawn.

> **TRY IT**
> Repeat each gesture five times, slowly at first, then without looking at the dial:

Gesture	Does
Ctrl+Alt+7, up	Zoom extents all
Ctrl+Alt+7, down	Isolate the selection (again = end isolate)
Alt+3, up / down	Select all / select none
Ctrl+Alt+1, up / right	Move / scale tool
Alt+7, up	Undo one step

RESULT

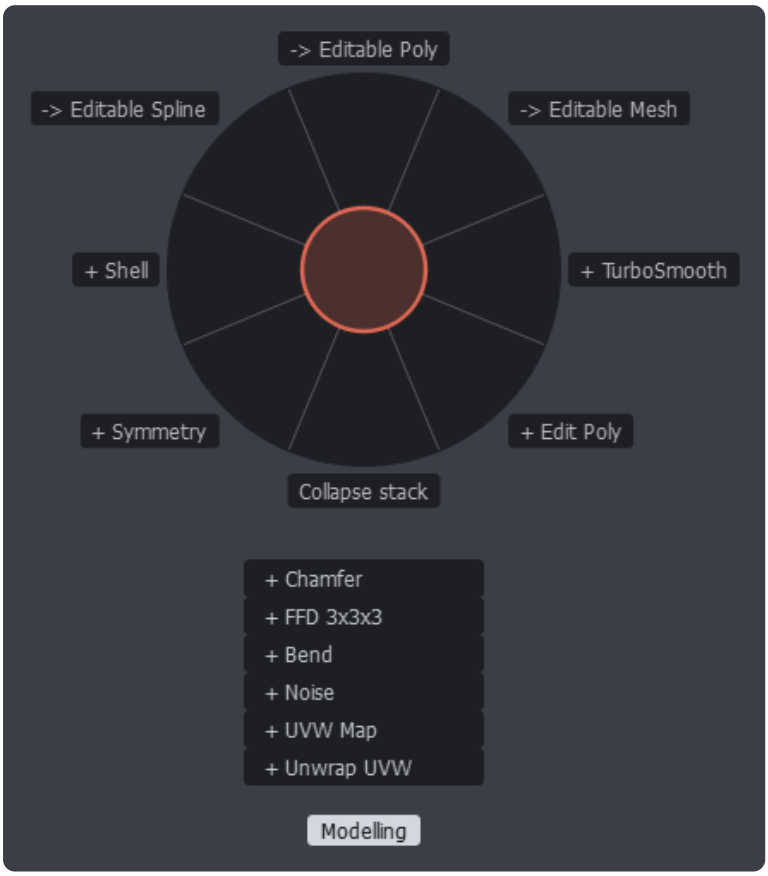
After a day these five need no looking. That is the whole idea - the rest of the dials follow the same way.

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4 More than eight - the list and the settings circle

3 min · You will learn: the rows under a dial, and a command's settings window

Some dials show **rows under the ring**: commands you use, but not often enough to spend a direction on. Keep holding the key, move down to a row and click it.

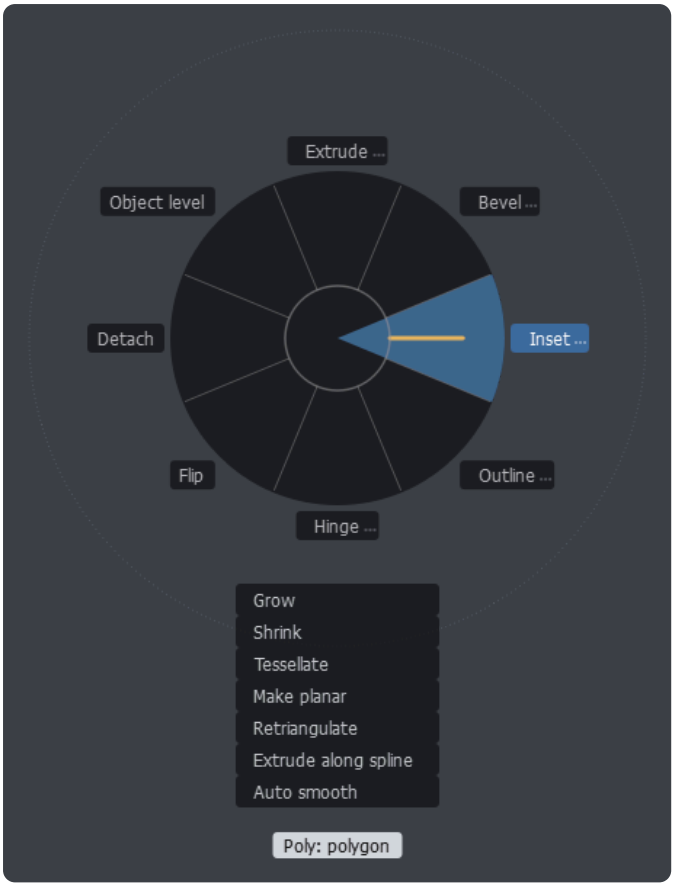


Alt+1 on an ordinary object: eight directions plus six rows (Chamfer, FFD, Bend, Noise, UVW Map, Unwrap UVW).

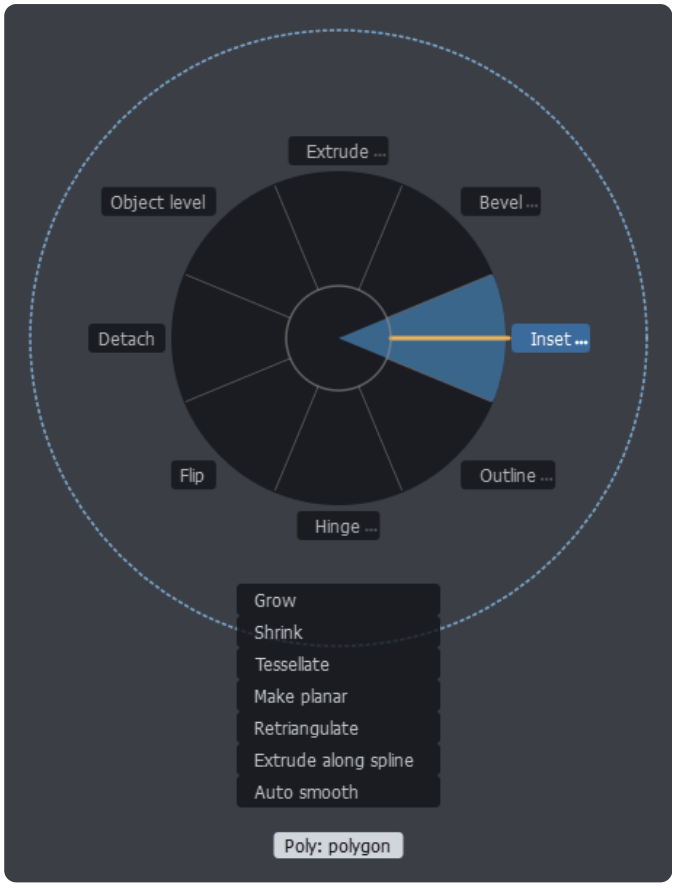
Commands that have a settings window (Chamfer, Extrude, Inset...) show **...** on their tile. A normal release runs the command **at once, without** that window - exactly as if you clicked its button in the Modify panel. For Inset, Extrude or Bevel that means: drag in the viewport to set the amount. Move **far out past the dial** before you let go and the command **opens its settings window** first, just like the small settings icon in a quad menu.

Where "far" starts is drawn on the dial: the **dotted circle** around it. Cross it and the circle **lights up** and the ... brightens - the window will open. Move back inside before you let go and the command runs without it after all.

A **flick** (tutorial 3) works the same way, even though the dial is not drawn: a short flick runs the command at once, a long one - longer than the distance to the dotted circle - opens the window. Your hand learns one movement, and it means the same thing fast or slow.



Inside the dotted circle: Inset runs at once.



Past it: the circle lights up - Inset opens its settings.

TRY IT

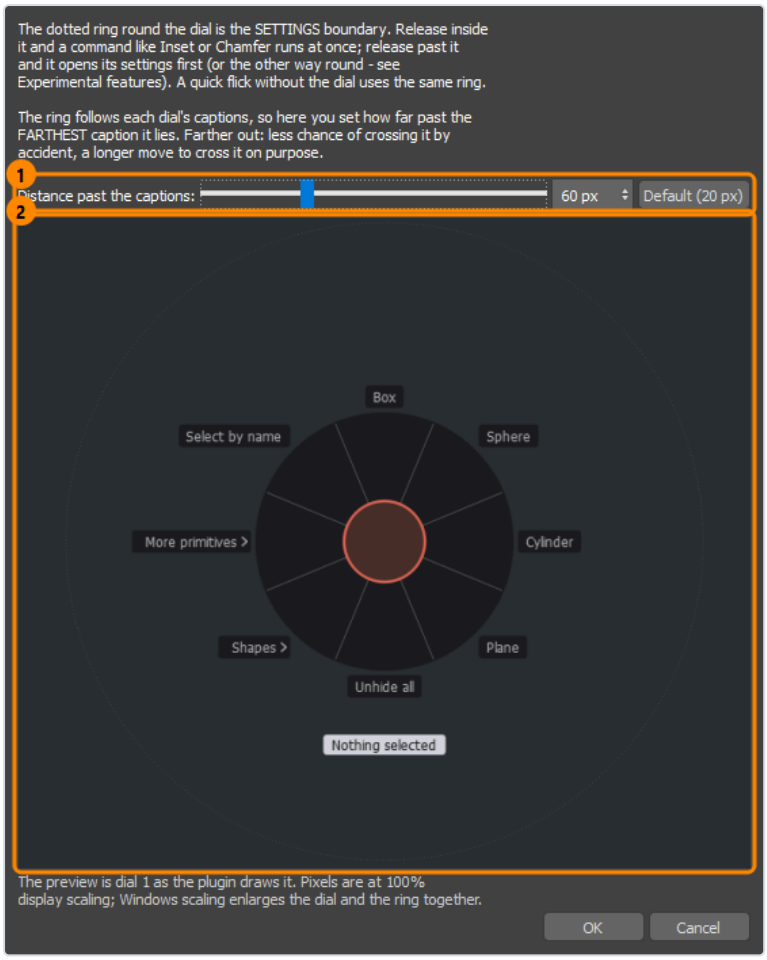
On a polygon, hold Alt+1, aim east (Inset) and release: Inset starts - drag in the viewport to inset. Do it again, but move far out before releasing: the Inset settings open this time. Then flick: a short flick east runs Inset at once, a long one opens the window.

Want it the other way round for a command you always set up first (the window unless you go far)? Select its direction in the editor and press **Settings...** (tutorial 7 shows where). For every command at once: **Experimental features**, the tick box **An ordinary release shows the action's SETTINGS**.

Move the dotted circle to suit your hand

The circle follows the captions of each dial, so you do not set its size - you set **how far past the farthest caption** it lies. Too close and your hand crosses it by accident; too far and crossing it on purpose is a long trip.

1. In the editor, press **Settings ring...** in the bottom bar.
2. Drag the slider or type a number: 0 to 200 px, 20 px by default. The picture is the real dial, drawn by the plugin with the circle where you put it. **Default (20 px)** puts it back.
3. OK, then **Save**. One number for every dial - and flicks follow the same circle.



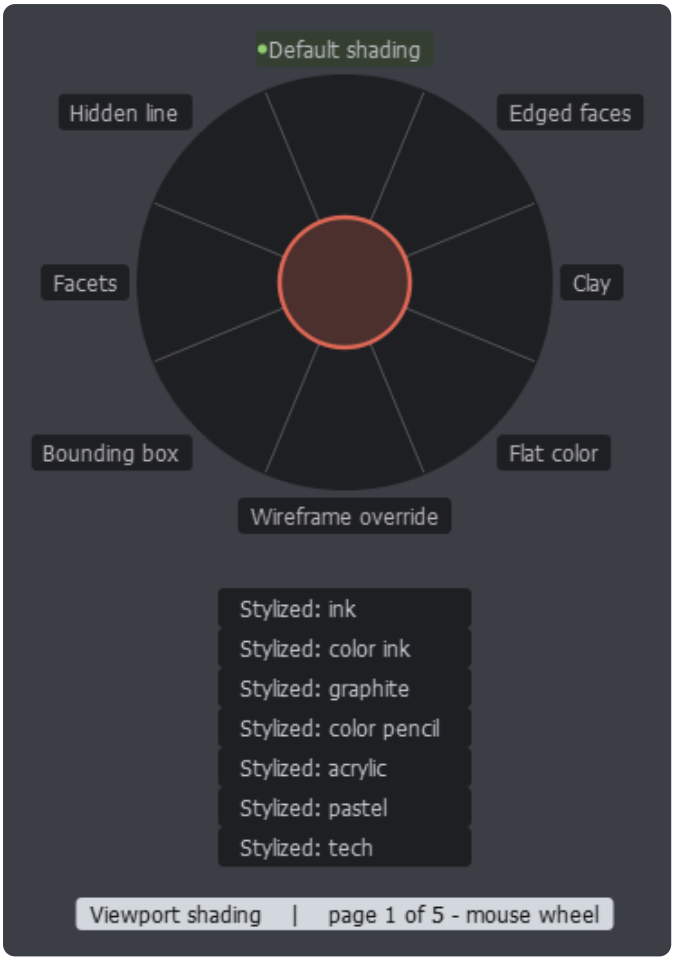
Settings ring...: the distance (1) and the dial with the circle where it will be (2).

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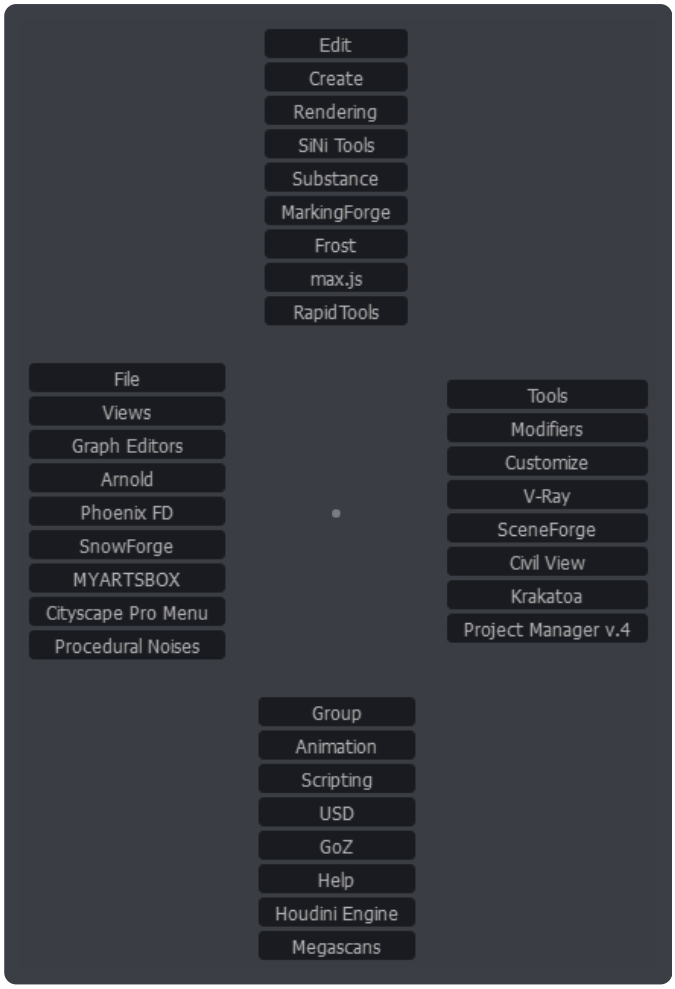
5 Undo by name, repeat, the viewport and the hotbox

4 min · You will learn: four dials that save you trips to the menus

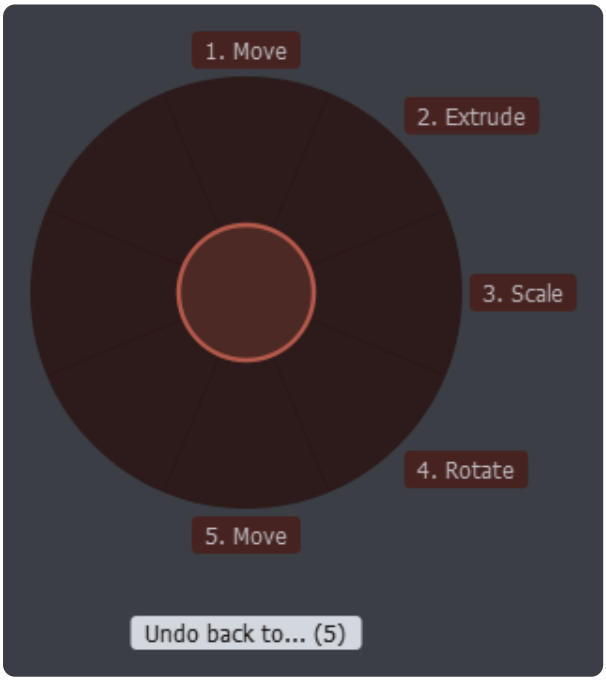
1. **Alt+7** - the undo dial shows your last eight operations by name. North undoes one, north-east two, and so on: undo three steps in one move.
2. **Alt+8** - recent commands: what you ran from the dials, newest first. Repeat the last one with a flick up.
3. **Alt+5** - the viewport: how it is drawn - default shading, clay, facets, wireframe and more. Keep holding the key and turn the **mouse wheel towards you** for the next pages: lighting and shadows, what is shown (grid, safe frames, statistics, ViewCube...) and the views (top, front, perspective...). A small dot marks a switch that is on. Rarer options sit in the rows under each page.
4. Hold **Alt+5** and turn the wheel one notch **away** from you instead: the **hotbox** - every 3ds Max menu as tiles, menus of other plugins included. Point at a title, then at an entry, and release - or click the entry. Esc or a right click closes it without running anything.



Alt+5 - how the viewport is drawn; the wheel turns to more pages.



Alt+5, one wheel notch away - the hotbox.



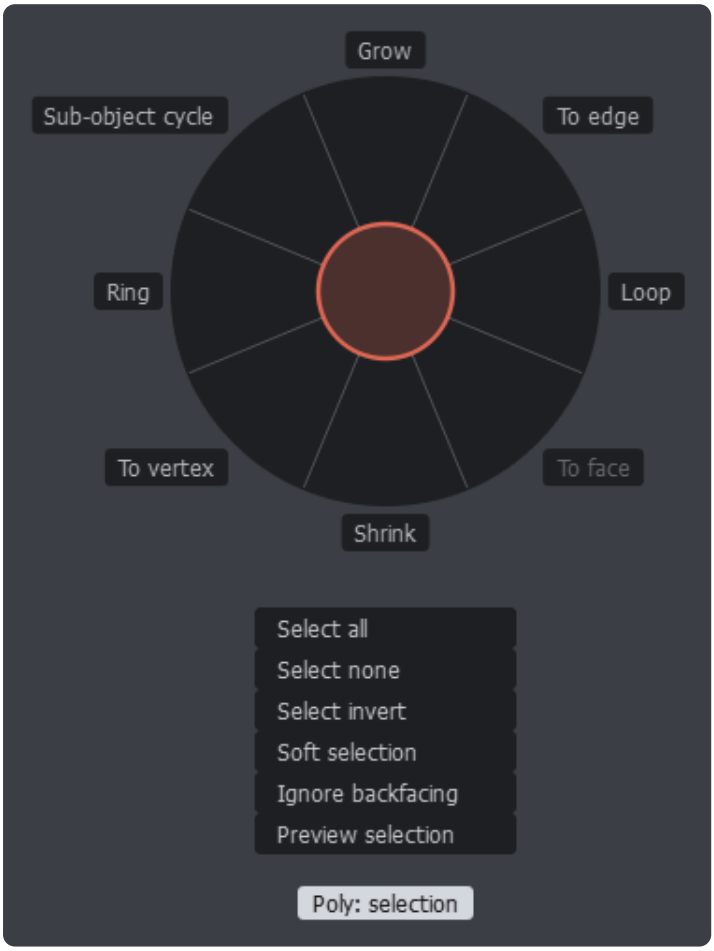
Alt+7 - undo by name.

> The **modifier stack** dial (Alt+6) fills itself too: the top eight modifiers of the selected object - aim at one to open it in the Modify panel.

6 Selection and hiding on sub-objects

2 min · You will learn: that other dials follow the sub-object level too

1. At polygon level hold **Alt+3** (Selection): Grow and Shrink face each other, Loop and Ring face each other; on an Editable Poly three corners convert the selection to vertices, edges or faces (under an Edit Poly modifier they are Select all, Select invert and Select none).
2. Hold **Ctrl+Alt+6** (Show and hide) at the same level: hide or unhide polygons, not whole objects.
3. Return to object level (Alt+1, north-west) and hold the same keys again - they work on whole objects.



Alt+3 at a sub-object level.

Five tricks worth knowing

Tricks 2-4 work straight away. Tricks 1 and 5 have to be switched on first in **MarkingForge > Experimental features** - where any of the five can also be switched off.

1. Several commands in one gesture (switch it on first)

1. Switch on **Multiple picks in one gesture** in **MarkingForge > Experimental features**.
2. Hold the key and move out **past the dial** towards a command.
3. Without letting go, **come back to the dashed circle** near the centre: the command is put in a queue, shown as a numbered row above the dial (*1 Ring > 2 Connect*).
4. Add more the same way, then let go inside the dashed circle (it turns green): they all run in order, and **one Ctrl+Z** undoes them all.
5. Changed your mind? Let go at the very centre or over an empty direction - nothing runs.

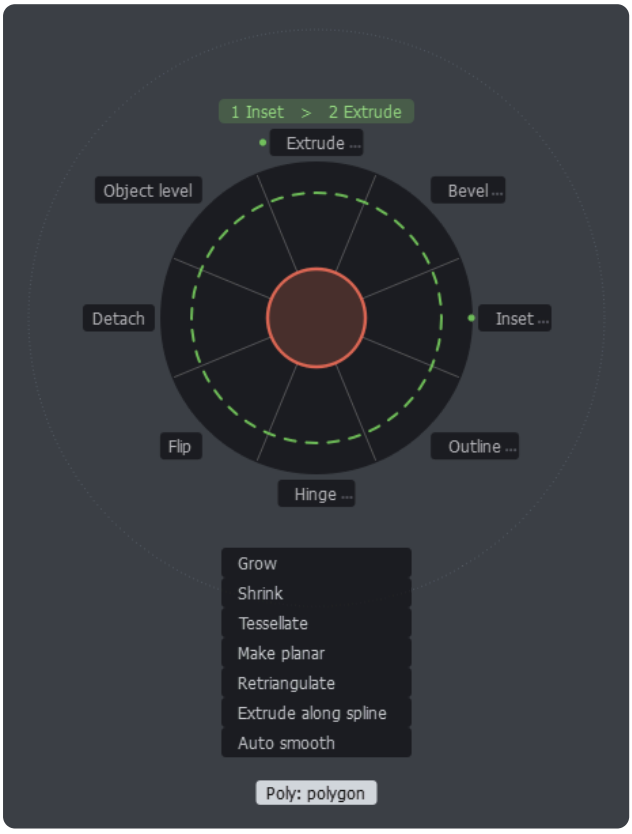


A command waits in the queue (the row above the dial). The dashed circle round the centre is where "back" is.

What chains well: commands that finish at once - selections (Loop, Ring, Grow), Flip, Connect, Weld, Cap, adding a modifier, display and view commands. A command that waits for you to **drag** in the viewport - Inset, Extrude, Bevel, Chamfer, Outline - stops the queue: drag, then **right-click**, and the next queued command starts by itself (3ds Max's status line names it; Q, W, E or R ends the tool too). Each such step is its own Ctrl+Z; another tool, another object or a new gesture drops the rest.

> Example: at edge level (press 2), select one edge and hold Alt+1 - out to **Ring** and back, out to **Connect** and back, then let go in the dashed circle: a new edge loop runs round the ring. One Ctrl+Z takes both back. A chain you repeat can become one direction: **MarkingForge > Save last chain as an item...**

A queue that waits for you. At polygon level (press 4), select a polygon and queue **Inset** (east) and **Extrude** (north), then let go in the dashed circle. Inset is on: drag, then **right-click** - Extrude starts by itself: drag, right-click. Select, Move, Rotate or Scale (Q, W, E, R) ends a tool the same way; Esc keeps it on; pan and zoom do not disturb the queue; another tool, another selected object or a new gesture drops the rest. Each waiting step is its own Ctrl+Z.



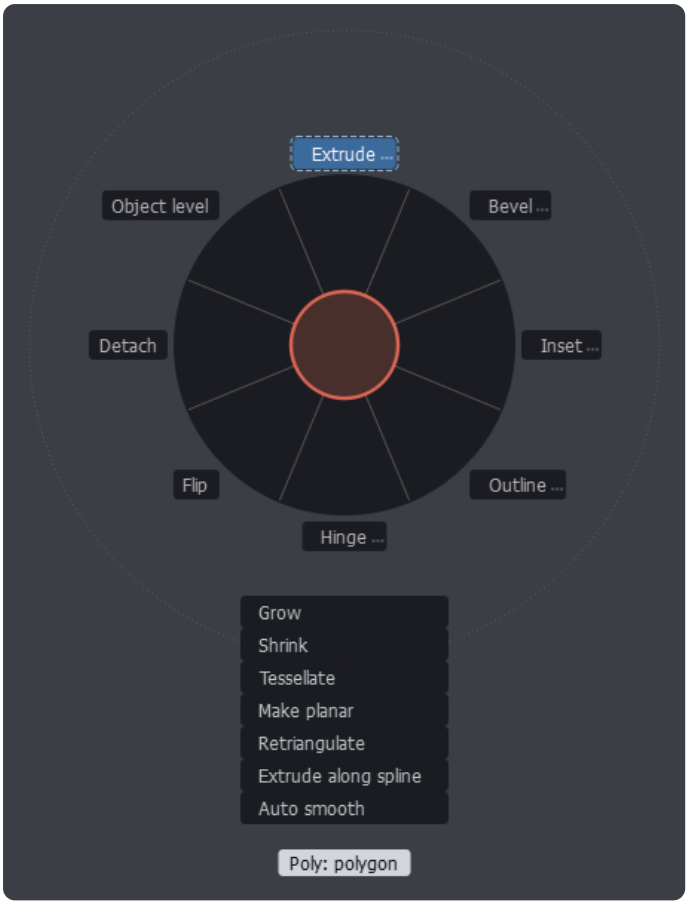
Inset (1) and Extrude (2) queued at polygon level; each waits for your drag and a right-click.

2. The dial of the object under the cursor

The dial looks first at the object **under the cursor**, and at the selection only when the cursor is over empty space. Point at a spline and open Alt+1: you get spline tools even while a box is selected, and the command works on that spline - MarkingForge selects it for you, and Ctrl+Z undoes both. Pointing at one of several selected objects keeps them all selected. Alt+6 shows the modifiers of the object you point at in the same way. A flick uses the object under the cursor at the moment you press the key. The hotbox and the search by typing never change the selection.

3. The suggested next command

MarkingForge notices pairs you repeat - Inset followed by Extrude, say. Once a pair has happened twice, the command you usually run next is **outlined** when the dial opens, and a **tap** of the key (press and let go without moving) runs it - unless you gave the dial a tap command of its own. It is only a suggestion: aim anywhere else as usual, nothing on the dial moves. It forgets when 3ds Max closes.



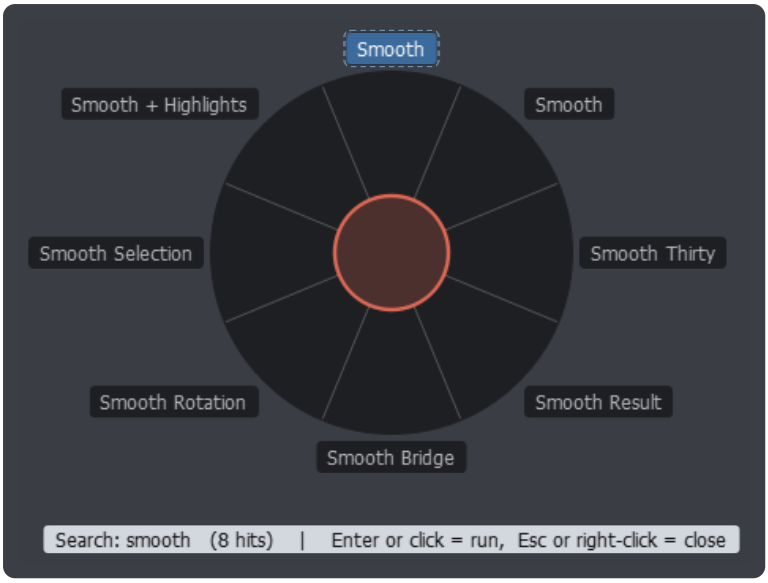
After Inset, Extrude is the outlined tile - a tap runs it.

4. Hints you can see

On the dial	Means
Dotted circle around the dial	The settings circle (tutorial 4) - it lights up once you are past it.
Dashed circle near the centre	A queue is waiting: come back inside it to add the command (trick 1).
Dashed outline of a direction	That direction is empty - letting go there cancels.
> after a caption	A submenu: move through it without letting go - the dial shows its contents in the same place (tutorials 1 and 13).
The circle in the centre	Let go inside it to cancel - always.
Outlined tile	What a tap runs - the suggested next command (trick 3).
A small dot before a caption	That switch is on (shadows, grid, edged faces...).
page 2 of 3 - mouse wheel under the dial	The dial has pages: keep holding the key and turn the mouse wheel (tutorial 20).
Nothing selected > More primitives under the dial	You are inside a submenu - the line says which one.

5. Search by typing (switch it on first)

1. Switch on **MarkingForge > Experimental features > Type-to-search**.
2. Hold a dial key and **type letters** - for example `smooth`. The dial becomes a search over every 3ds Max command and **stays open**, so you can let go of the key.
3. Aim at a result, or press **Enter** for the best one. **Backspace** deletes a letter, **Esc** closes the search.



Alt+1, then `smooth`: commands with that name, the best match outlined.

> Typing works with the **Alt** and **Shift+Alt** dials. With **Ctrl** held, letters stay 3ds Max shortcuts (Ctrl+S, Ctrl+Z...), so the Ctrl+Alt dials have no search.

The key map

Digits 1-8 with **Alt**, **Ctrl+Alt** and **Shift+Alt**. The digit is the place, Alt / Ctrl+Alt / Shift+Alt the row - count from a key you know.

Key	Dial	Key	Dial	Key	Dial
Alt+1	Modelling	**Ctrl+Alt+1**	Transform	**Shift+Alt+1**	Viewport display
Alt+2	Modifiers	**Ctrl+Alt+2**	Align and pivot	**Shift+Alt+2**	Selection sets
Alt+3	Selection	**Ctrl+Alt+3**	Lights and cameras	**Shift+Alt+3**	Materials
Alt+4	Create	**Ctrl+Alt+4**	UV mapping	**Shift+Alt+4**	Render
Alt+5	Viewport shading	**Ctrl+Alt+5**	Snaps and grid	**Shift+Alt+5**	Animation
Alt+6	Modifier stack	**Ctrl+Alt+6**	Show and hide	**Shift+Alt+6**	Scene
Alt+7	Undo	**Ctrl+Alt+7**	View	**Shift+Alt+7**	Tools and setup
Alt+8	Recent commands	**Ctrl+Alt+8**	Viewports	**Shift+Alt+8**	File

Print the cheat sheet and keep it beside the keyboard for the first week: **MarkingForge > Dial cheat sheet (HTML + PDF)...** - your dials as small squares with their keys, sorted by key.

MarkingForge - dial cheat sheet

2026-10-07 13:41 · 24 dials and 4 more pages · sorted by shortcut, because that is what the hand presses

Alt+1 Modelling

slot 1

1. when window under the cursor matches slate

↖ NW Pick from selected	↑ N New material	↗ NE Edit selected's material
← W Get from object	Materials (Slate)	→ E Assign to selection
↙ SW Material Explorer	↓ S Material/Map Browser	↘ SE Show in viewport

2. when window under the cursor matches trackView

↖ NW Curve mode	↑ N Tangents: auto	↗ NE Tangents: smooth
← W Isolate curve	Track View	→ E Tangents: linear
↙ SW Frame all	↓ S Flatten	↘ SE Tangents: stepped

3. when window under the cursor matches particleView

↖ NW Rename	↑ N Move up	↗ NE Copy
← W Move left	Particle View	→ E Move right
↙ SW Cut	↓ S Move down	↘ SE Paste

4. when window under the cursor matches sceneExplorer

↖ NW Zoom to selected	↑ N Select all	↗ NE Select children
← W Unhide all	Scene Explorer	→ E Hide
↙ SW Unfreeze all	↓ S Isolate	↘ SE Freeze

List below the dial

1. Expand all
2. Collapse all
3. Find selected
4. Rename

5. when window under the cursor matches commandPanel

Modifier stack. Eight modifiers FROM THE TOP of the selected object's stack. Requires EXACTLY ONE selected object - otherwise the dial opens empty and says so in its footer.

6. when modifier = Unwrap_UVW

↖ NW Stitch	↑ N Quick relax	↗ NE Pack
← W Weld selected	UV (Unwrap)	→ E Edge loop
↙ SW Break	↓ S Straighten	↘ SE Edge ring

List below the dial

1. Open UV Editor

7. when modifier = Edit_Poly and subobject: 1 - usually vertex

↖ NW Object level	↑ N Chamfer	↗ NE Weld
← W Target Weld	Edit Poly: vertex	→ E Connect
↙ SW Remove	↓ S Extrude	↘ SE Break

List below the dial

1. Grow
2. Shrink
3. Remove isolated

8. when modifier = Edit_Poly and subobject: 2 - usually edge

↖ NW Object level	↑ N Connect	↗ NE Chamfer
← W Remove	Edit Poly: edge	→ E Loop
↙ SW Weld	↓ S Extrude	↘ SE Ring

List below the dial

1. Split
2. Turn
3. Grow
4. Shrink

9. when modifier = Edit_Poly and subobject: 3 - usually border / spline

↖ NW Object level	↑ N Cap	↗ NE Chamfer
← W Grow	Edit Poly: border	→ E Extrude
↙ SW Shrink	↓ S Create shape	↘ SE Connect

List below the dial

1. Bridge

The printable cheat sheet.

PART 2

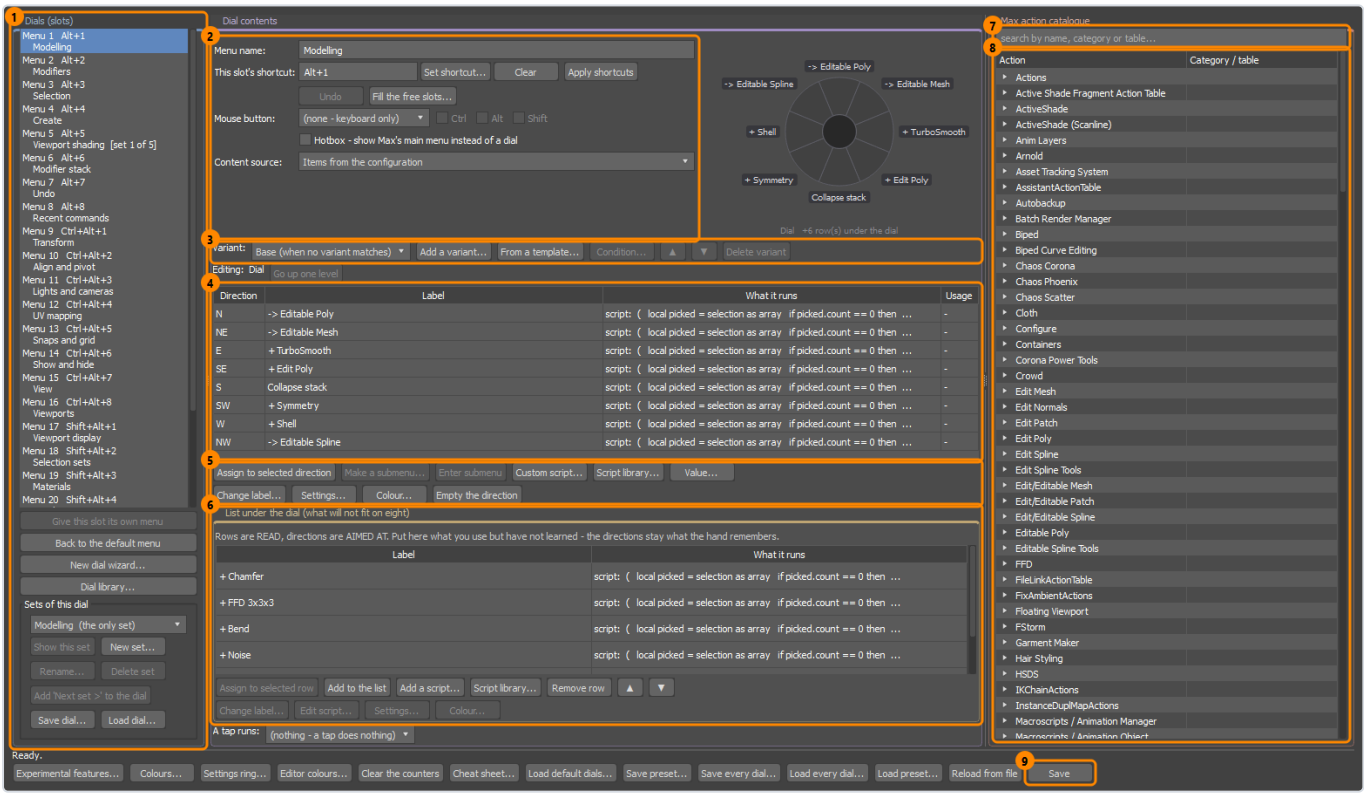
Making it yours

Sixteen tutorials in the menu editor. Nothing you change reaches the dials until you press **Save** - so try things freely; **Reload from file** throws away what you have not saved. One exception: the editor's own colours (tutorial 17) are kept as soon as you press OK.

7 A tour of the editor

2 min · You will learn: where everything is

Open **MarkingForge > Menu editor...**



The menu editor.

1. **Dials** - all 24 dials and their keys. Click one to edit it; right-click it to save or load it.
2. **Dial contents** - its name, key, mouse button, and where its contents come from.
3. **Variant** - versions of the dial for different situations, and when each one shows.
4. **Directions** - N to NW: the caption, what runs, how often you used it.
5. **Direction tools** - submenu, script, script library, value (a number slider), caption, settings, colour, empty.
6. **List under the dial** - extra rows.
7. **Search** - find any 3ds Max command by name.

8. **Catalogue** - every 3ds Max command (over 4000), sorted into groups.
9. **Save** - writes your dials; 3ds Max picks them up at once.

Next to the dial contents the editor **draws the dial** you are editing - click a wedge to select that direction. Under the list of dials: **New dial wizard...**, **Dial library...** and **Sets of this dial**; a dial with several pages shows them as **tabs** above its contents (tutorials 20-22). The bottom bar has what concerns all dials: colours, the cheat sheet, presets, and saving or loading every dial (tutorials 16-18).

> **Closing the editor never loses your work by surprise.** With unsaved changes, Esc or the window's X asks: **Save**, **Discard** or **Cancel**.

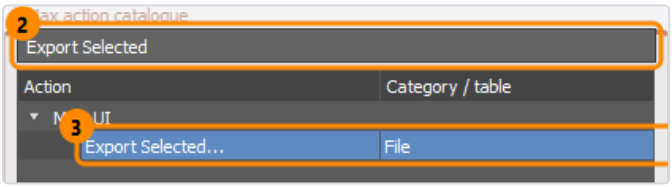
---

8 Put a command on a direction

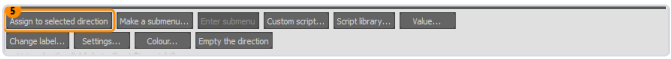
3 min · You will learn: the one skill everything else builds on

Example: on the **File** dial (Shift+Alt+8) replace *Save copy as* (south-east) with **Export Selected**.

1. In the dial list pick **Menu 24** (Shift+Alt+8, File).
2. Type `Export Selected` into the search field.
3. Click **Export Selected...** in the catalogue.
4. Click the **SE** row in the directions table.
5. Press **Assign to selected direction** under the table - or double-click the command, or drag it onto the row.
6. Press **Save**.



Steps 2 and 3: type, then click the command.



Step 5: the button under the directions table.

Direction	Label	What it runs	Usage
N	Save	action: Main UI	-
NE	Save as...	action: Main UI	-
	Save incremental	action: Main UI	-
SE	Save copy as...	action: Main UI	-
S	Open...	action: Main UI	-
SW	Import...	action: Main UI	-
W	Merge...	action: Main UI	-
NW	Hold	action: Main UI	-

Step 4: the SE row selected - it still holds Save copy as.

Direction	Label	What it runs	Usage
N	Save	action: Main UI	-
NE	Save as...	action: Main UI	-
E	Save incremental	action: Main UI	-
SE	Export Selected...	action: Main UI (name from Max)	-
S	Open...	action: Main UI	-
SW	Import...	action: Main UI	-
W	Merge...	action: Main UI	-
NW	Hold	action: Main UI	-

After step 5: SE runs Export Selected. *(name from Max)* means the caption is 3ds Max's own - tutorial 9 shortens it.

TRY IT

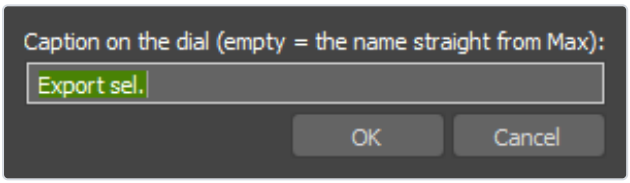
Hold Shift+Alt+8 in the viewport - south-east is Export Selected.

The same command can appear in several groups (Select All is in seven). The *Category* column tells them apart - for commands that work on the scene, pick *Main UI*.

9 A shorter caption

1 min · You will learn: to keep the dial readable

1. Select the direction.
2. Press **Change label...**
3. Type the caption and press OK. Leave it empty to use 3ds Max's own name.
4. Save.

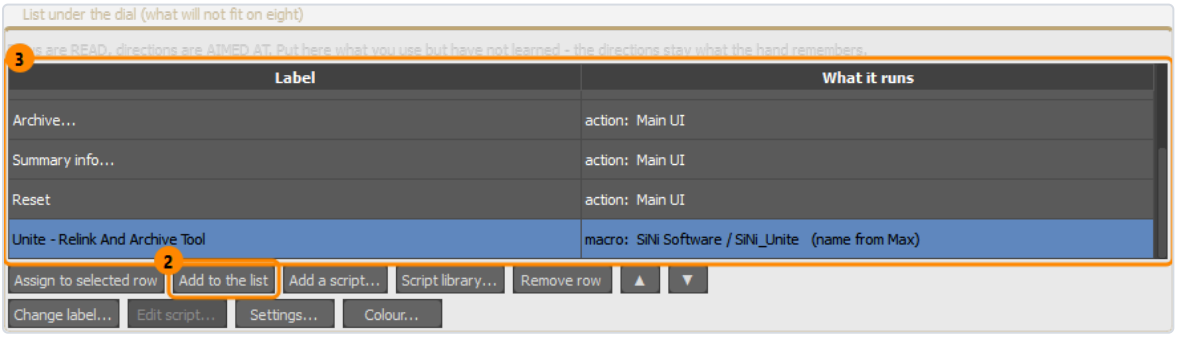


Two words beat a cut-off name on the dial.

10 Rows under the dial

2 min · You will learn: to keep more than eight commands on one key

1. Pick the dial.
2. Find the command in the catalogue.
3. Press **Add to the list** (or drag the command into the list).
4. Reorder with ▲ ▼, remove with **Remove row**. Up to twelve rows.
5. To edit a row, select it: **Change label...**, **Edit script...**, **Settings...** and **Colour...** sit in the list's own bar. **Add a script...** adds a row that runs a script of your own.
6. Save.



The list pane: the rows, and Add to the list under them.

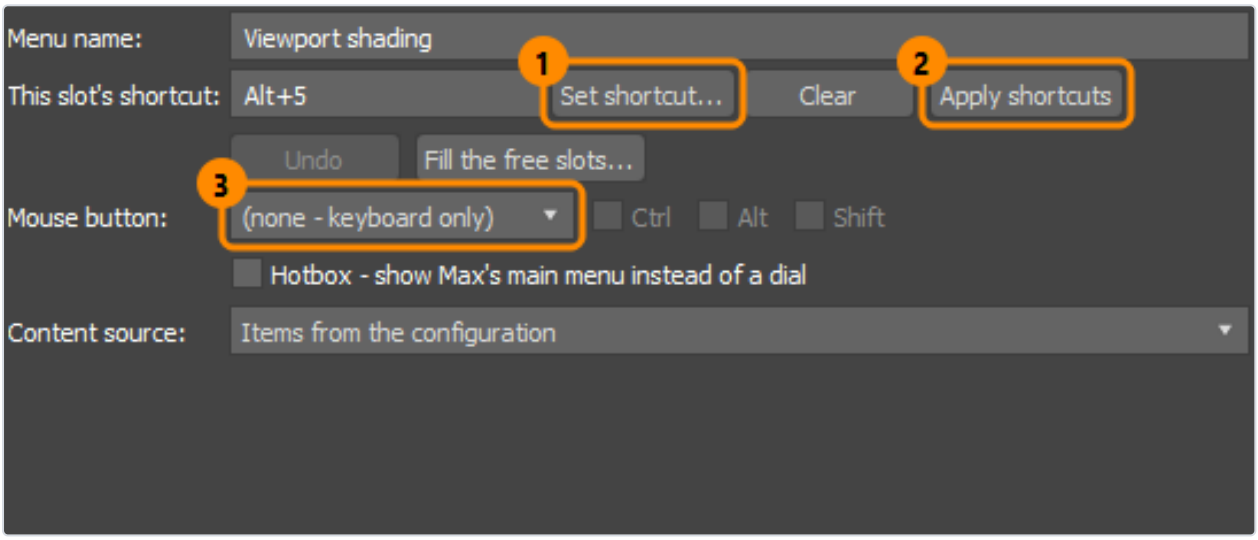
Put here what you use but have not learned. Promote a row to a direction once your hand wants it.

11

Change a dial's key, or open it with the mouse

2 min · You will learn: shortcuts and mouse buttons

1. Pick the dial.
2. Click **Set shortcut...**. A small window opens: press the new keys and click **Set**. A key that already opens another dial can be used - it moves to this dial. **Clear** removes the key.
3. Press **Apply shortcuts** - the keys go into 3ds Max; your other shortcuts stay as they are, and a backup is made. **Undo** next to it brings back the shortcuts from before; **Fill the free slots...** gives the standard key to every dial that has none.
4. Optional: pick a **Mouse button** together with at least one of Ctrl / Alt / Shift - for example Ctrl+Shift + *middle button*.
5. Save.



The dial settings: Set shortcut (1), Apply (2), mouse button (3).

The keys are also in 3ds Max's **Customize > Hotkey Editor** as *MarkingForge: Menu 1 ... 24*.

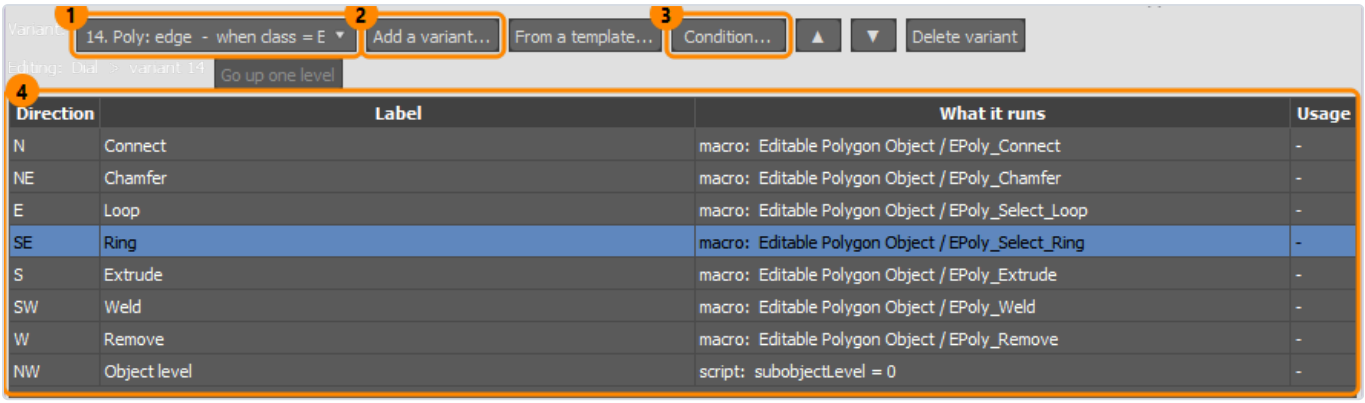
---

12 A variant - same key, other contents in another situation

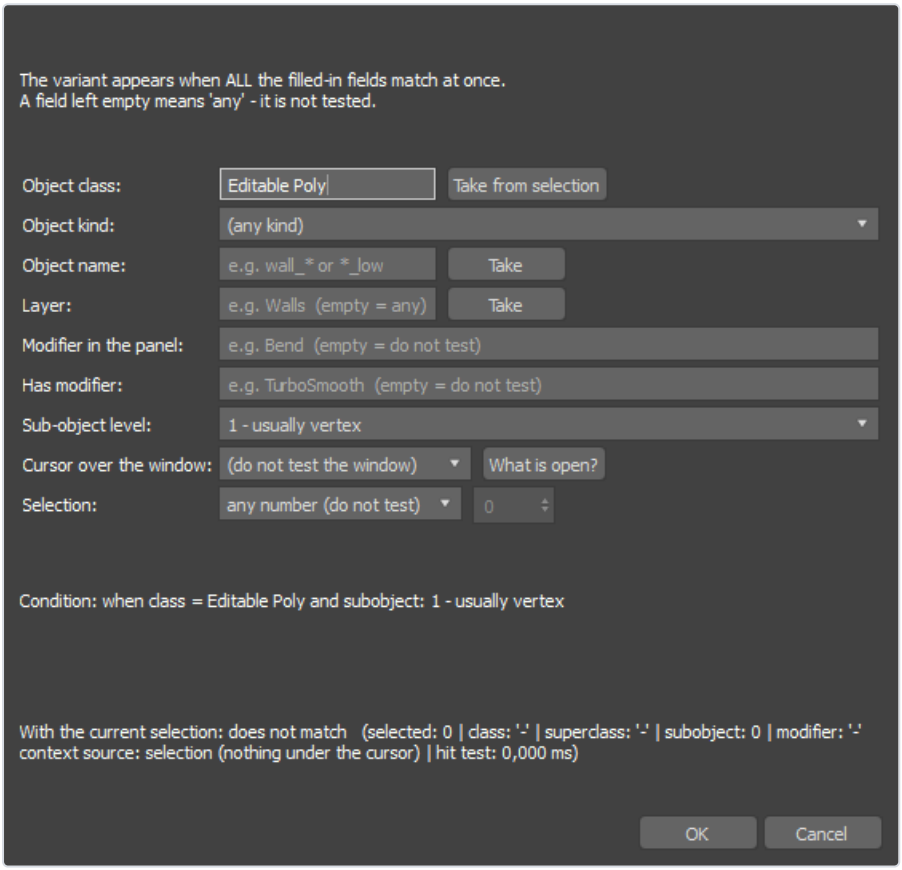
4 min · You will learn: to make one key do different things in different situations

Example: give a dial different contents when you work on the edges of an Editable Poly.

1. Pick the dial and press **Add a variant...** - it starts as a copy of what you see.
2. In the window that opens, select an Editable Poly in 3ds Max, go to edge level and press **Take from selection**: the object type and the level fill in.
3. OK. The **Variant** list now shows your variant - fill its directions like any dial.
4. If two variants fit, the higher one wins - keep the more specific ones on top (▲ ▼).
5. Save.



The Modelling dial's built-in edge variant: the variant list (1), Add a variant (2), Condition (3) and the variant's own directions (4).



The condition window (here: Editable Poly, vertex level). Empty fields are not tested.

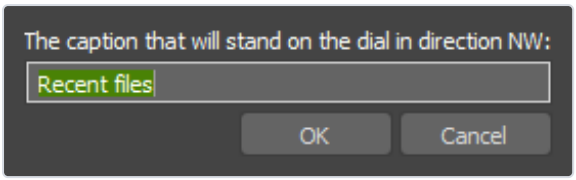
> A variant can also depend on the **window under the cursor** (Slate, Track View, Scene Explorer...), the modifier open in the Modify panel, how many objects are selected, an object's name or its layer.

---

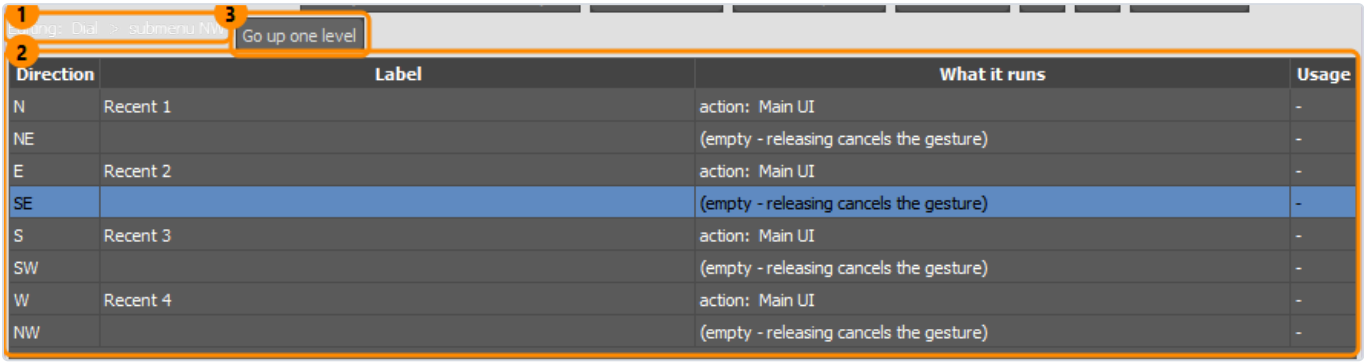
13 A submenu

3 min · You will learn: to put a whole dial behind one direction

1. Select a direction and press **Make a submenu...**
2. Type its caption - it stands on the dial.
3. You are now inside the submenu (the line above says *Editing: Dial > submenu NW*). Fill its directions.
4. Press **Go up one level** to return. Save.



Step 2: the caption for the submenu.



Step 3: inside the submenu - the path (1), its own directions (2), and the way back (3).

TRY IT

On the dial, move through the submenu's direction without letting go: the dial stays where it is and shows the submenu, and the line under it says where you are (*Nothing selected* > *More primitives*). Move back inside the dial to pick; move into the centre to go back.

Submenus in the built-in dials. Open them to see what a submenu is good for - groups of commands that would not fit on eight directions:

Dial	Direction	Submenu
Alt+4 Create	S, SW, W, NW	Shapes, More primitives, Helpers, Cameras and lights
Alt+2 Modifiers	SW, NW	Deform (Bend, Twist, Taper, FFD...), Geometry (Mirror, Array, Cap Holes, OpenSubdiv...)
Alt+1 Modelling, nothing selected	SW, W	Shapes, More primitives

In the editor's drawing of the dial a submenu is marked with >; **double-click** it (or its row in the table) to step inside.

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14 Scripts on a dial - ready-made or your own

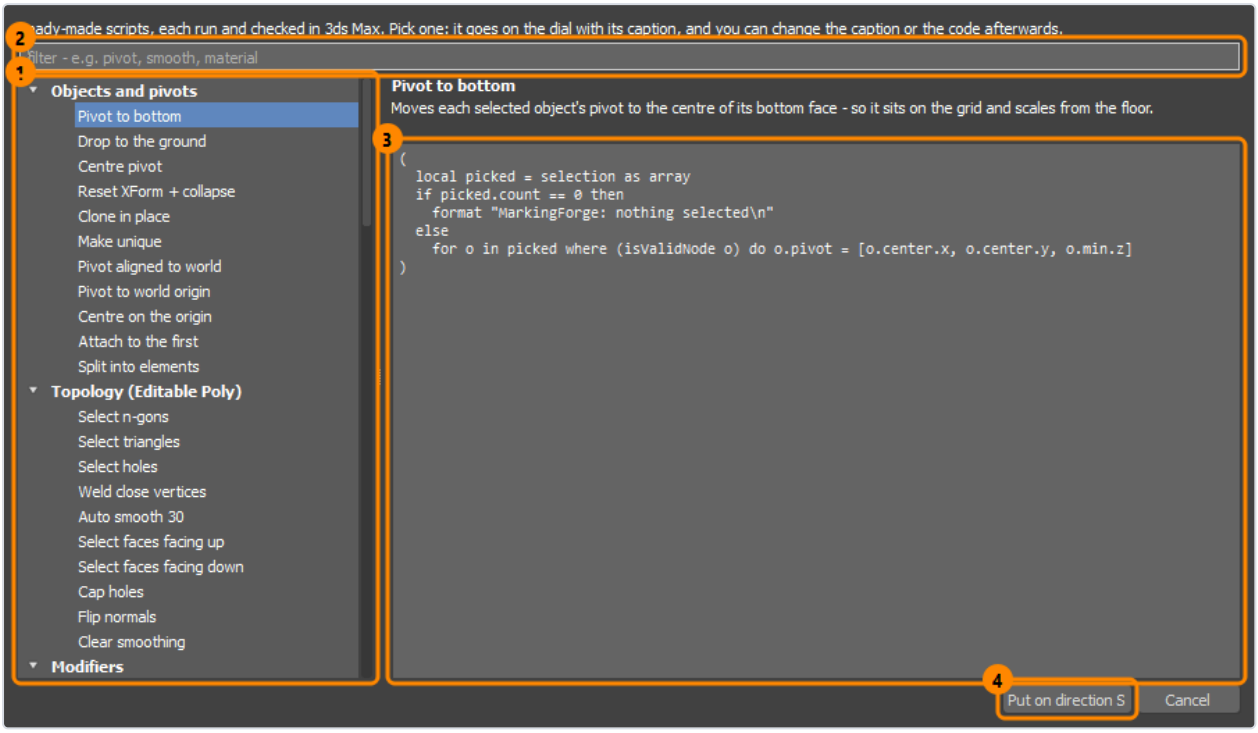
6 min · You will learn: to run a script from a gesture, with or without writing one

A direction or a list row can run a **script** - a small program for 3ds Max. You do not have to write one: the **Script library** has 102 ready-made scripts in eleven groups - objects and pivots, topology, modifiers, advanced modelling (loops and rings, chamfer, bridge, bevel, planarize, symmetrize, greeble...), UV mapping, materials and display, selection and transforms, layout and placement, shapes and splines, scene and layers, cameras, lights and render. Every one was tried in 3ds Max before it went into the list.

A. From the library - no code

1. Pick the dial and select a direction - or, to keep your directions, use the list under the dial.
2. Press **Script library...** (under the directions, or in the list's bar).

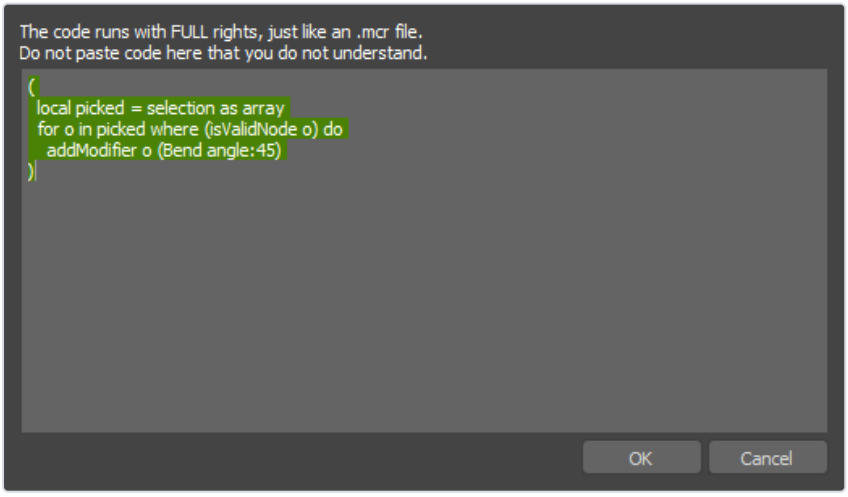
3. Click a script on the left: the right side says what it does (and shows its code, if you are curious). Type in the filter to narrow the list - *pivot*, *smooth*, *material*.
4. Press **Put on direction S** - the button names the direction you selected - (or **Add to the list**), then **Save**.



The script library - groups on the left, what the script does and its code on the right.

B. A script of your own

1. Select a direction and press **Custom script...** (it reads **Edit script...** once the direction holds a script).
2. Paste the script (MAXScript) and press OK.
3. Give it a caption with **Change label...**
4. Save.



A script that adds a Bend to every selected object.

Four scripts every modeller uses

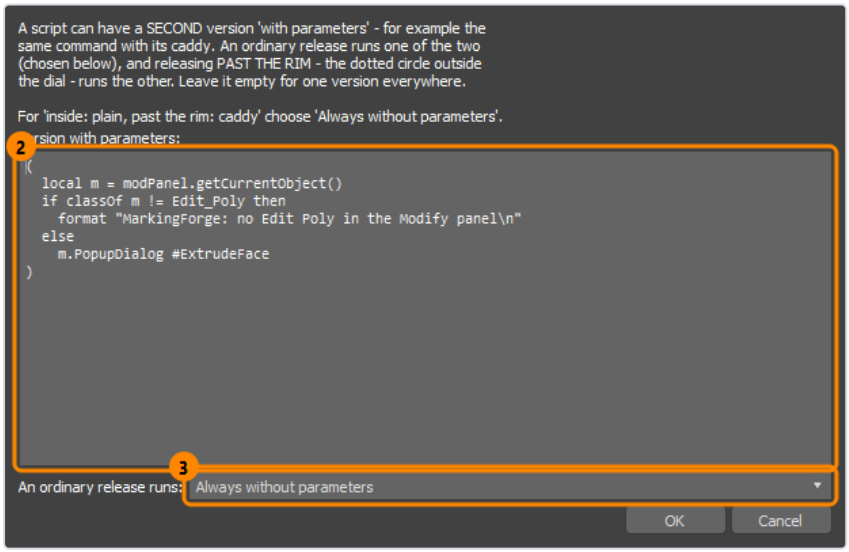
All four are in the library. Put them on the list under the Modelling dial (**Menu 1**, Alt+1) and they are one gesture away whatever you model.

Script	Try it like this
Pivot to bottom	Create a sphere, hold Alt+1, pick the row: the pivot jumps to the bottom of the sphere. Pair it with **Drop to the ground** and objects stand on the grid.
Select n-gons	Convert a cylinder to Editable Poly, pick the row: both caps light up at polygon level, and the Listener window says how many were found. Fix them before smoothing.
Reset XForm + collapse	Scale a box to 300 % and rotate it, then pick the row: the scale and rotation are baked into the mesh and the object is clean - what Shell, Symmetry and exporters expect.
Smoothing on/off	Give a few objects TurboSmooth, select them all, pick the row: every TurboSmooth switches off at once; pick it again and they come back.

1. Select **Menu 1** in the editor's left pane.
2. In the list bar press **Script library...**, pick **Pivot to bottom**, press **Add to the list**.
3. Repeat for the other three.
4. Save, then hold Alt+1: the four rows sit under the dial.

C. Two versions of one script - plain, and with its settings window

1. Select a direction that runs a script and press **Settings...**
2. Paste the second version into **Version with parameters** - for Extrude on an Edit Poly modifier: `(modPanel.getCurrentObject()).PopupDialog #ExtrudeFace`
3. Choose **Always without parameters**: a normal release runs the plain version; letting go past the dotted circle runs the version with the settings window.
4. OK, then Save. In the table the script now reads *[+ with parameters]*.



Settings... on a script: the version with parameters and which one an ordinary release runs.

> The library's **Extrude (Edit Poly)** comes set up this way. A script can do anything 3ds Max can - only use scripts you trust. The Reference (chapter 8) lists every library script.

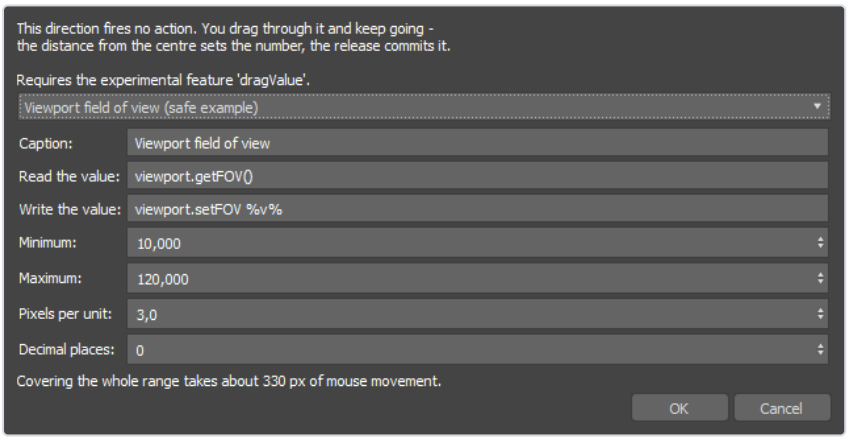
15 Values - five gesture sliders

10 min · You will learn: to set a number by how far you move, with five recipes

A **value** direction is a slider on the dial. Hold the key, move into that direction and keep going: the further you move, the bigger the number (you see it next to the caption), and when you let go the number is set in the scene. Moving back makes it smaller.

You do not have to write anything: the **Value...** window has ready-made examples, and picking one fills in every field. The five below are the first five in that list - the tables only show what you will see. The two code fields simply say where the number is read from and where it goes (`%v%` is the number).

1. Switch on **MarkingForge > Experimental features > Direction that sets a value** (in the editor: **Experimental features...** in the bottom bar). Without it the direction does nothing and the dial says so.
2. In the editor pick a dial, select an **empty direction** and press **Value...**
3. Pick an example from the list at the top of the window - every field fills in. Change the caption if you like.
4. Look at the line at the bottom of the window: it says how far you must move the mouse to go from the smallest number to the largest. About 150 to 500 pixels feels right; **Pixels per unit** (how far you move for one step) changes it.
5. **OK**, then **Save**. The dial in 3ds Max picks the change up at once.



The value window with the first example, the viewport field of view.

Example 1 - Viewport field of view

Zoom the lens of the perspective view without touching the camera - the safe one: needs no selection and changes only the view.

Field	Value
Caption	Viewport field of view
Read the value	`viewport.getFOV()`
Write the value	`viewport.setFOV %v%`
Minimum / Maximum	10 / 120
Pixels per unit	3
Decimal places	0

TRY IT

In a perspective view drag into the direction and on: the view widens as you move out, narrows as you come back; release at the angle you like.

Example 2 - Time slider frame

Scrub the animation from the dial - a frame per three pixels.

Field	Value
Caption	Time slider frame
Read the value	`sliderTime.frame`
Write the value	`sliderTime = %v%`
Minimum / Maximum	0 / 100
Pixels per unit	3
Decimal places	0

TRY IT

Move out slowly and watch the time slider follow; release on the frame you want.

Example 3 - TurboSmooth iterations of the selection

Raise or lower the smoothing of every selected object that has TurboSmooth - 40 pixels per step, so each step is easy to hit.

Field	Value
Caption	TurboSmooth iterations of the selection
Read the value	`selection[1].modifiers[#TurboSmooth].iterations`
Write the value	`for o in selection where o.modifiers[#TurboSmooth] != undefined do o.modifiers[#TurboSmooth].iterations = %v%`
Minimum / Maximum	0 / 4
Pixels per unit	40
Decimal places	0

TRY IT

Select a few objects with TurboSmooth; the number next to the caption steps 0, 1, 2, 3, 4 as you move.

Example 4 - Uniform scale of the selection

Scale every selected object evenly, from a tenth to five times, with two decimals.

Field	Value
Caption	Uniform scale of the selection
Read the value	`selection[1].scale.x`

Write the value	`for o in selection do o.scale = [%v%, %v%, %v%]`
Minimum / Maximum	0.1 / 5
Pixels per unit	60
Decimal places	2

TRY IT
Select objects, move out: they grow in the viewport only after the release - the number shows where you are while you move.

Example 5 - Thickness of the selected splines

Make splines renderable and set their thickness - a quick tube from a line.

Field	Value
Caption	Thickness of the selected splines
Read the value	`selection[1].render_thickness`
Write the value	`for o in selection where superClassOf o == shape do (o.render_renderable = true; o.render_displayRenderMesh = true; o.render_thickness = %v%)`
Minimum / Maximum	0.1 / 20
Pixels per unit	15
Decimal places	1

TRY IT
Draw a few lines or circles, select them and set the thickness in one move.

? next to the caption means there is nothing to read - usually nothing (or the wrong kind of object) is selected. Letting go then changes nothing. Examples 3-5 change **every** selected object and skip the ones they do not fit (no TurboSmooth, not a spline).

Changed your mind? Move into the centre of the dial and let go - nothing changes.

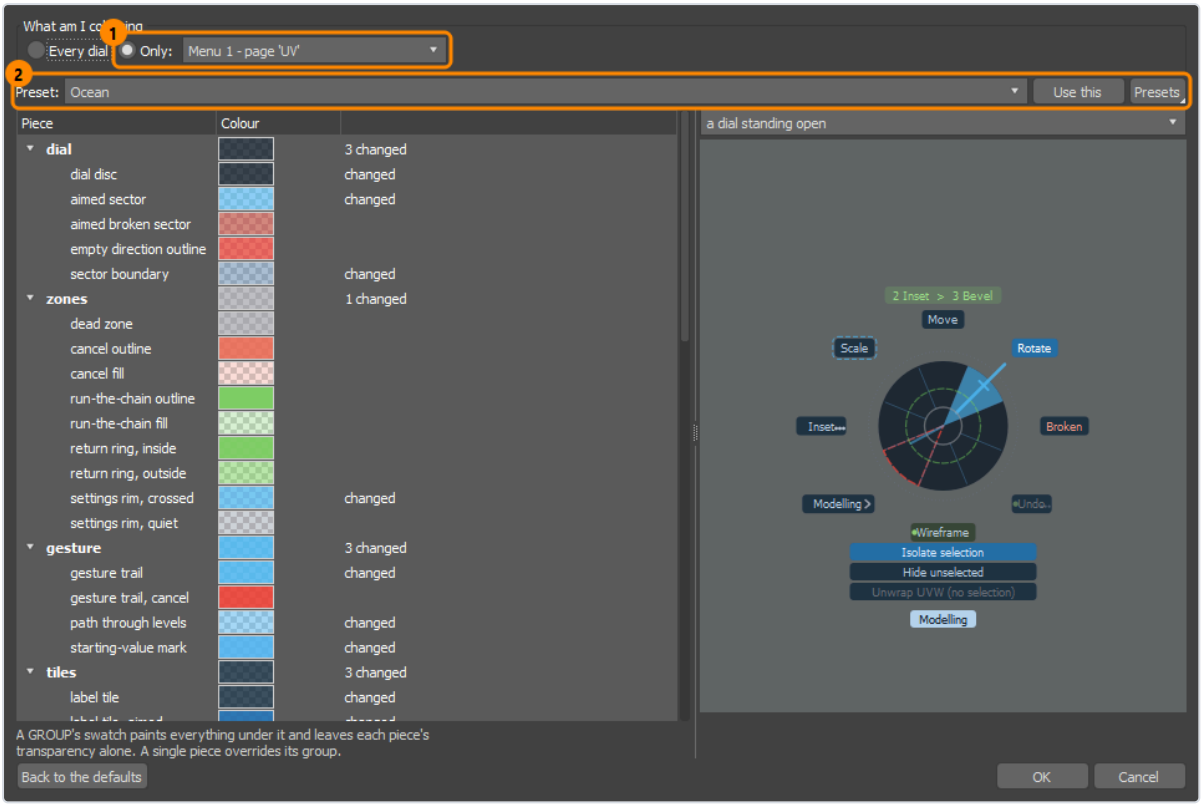
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16 Colours and ready-made looks

3 min · You will learn: to recognise a dial before reading it

1. Press **Colours...** in the bottom bar of the editor.
2. Under **What am I colouring** choose **Every dial**, or **Only** this one. For a dial with sets the **Only:** list also offers *Menu N - every page* and each page on its own (*Menu N - page 'UV'*) - the pages the mouse wheel turns (tutorial 20).
3. Pick a look from **Preset:** and press **Use this**. There are thirty ready-made looks - Graphite, Ocean, Ember, Neon, High contrast and more, five of them with light tiles - and the *MarkingForge default*.
4. Or click a group (dial, zones, tiles, text, list...) or a single piece and pick a colour. The list above the preview shows the moment you are colouring: the dial open, aiming at a direction, about to cancel, a queue waiting.

5 OK, then Save.



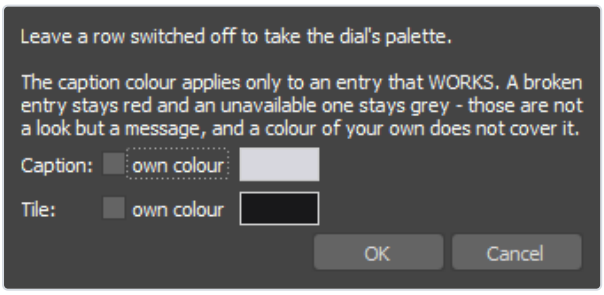
Colours...: what you colour - here one page of Menu 1 (1) - and the preset row (2).

> **Your own looks. Presets > Save these colours as a preset...** keeps your look under a name; **Export this preset to a file...** saves it to take to another computer, and **Import a preset from a file...** brings it in. A dial that uses the default menu has no colours of its own: press **Give this slot its own menu** first.



The thirty ready-made looks.

For one command only - a red caption on a destructive one, say - select its direction and press **Colour...**:



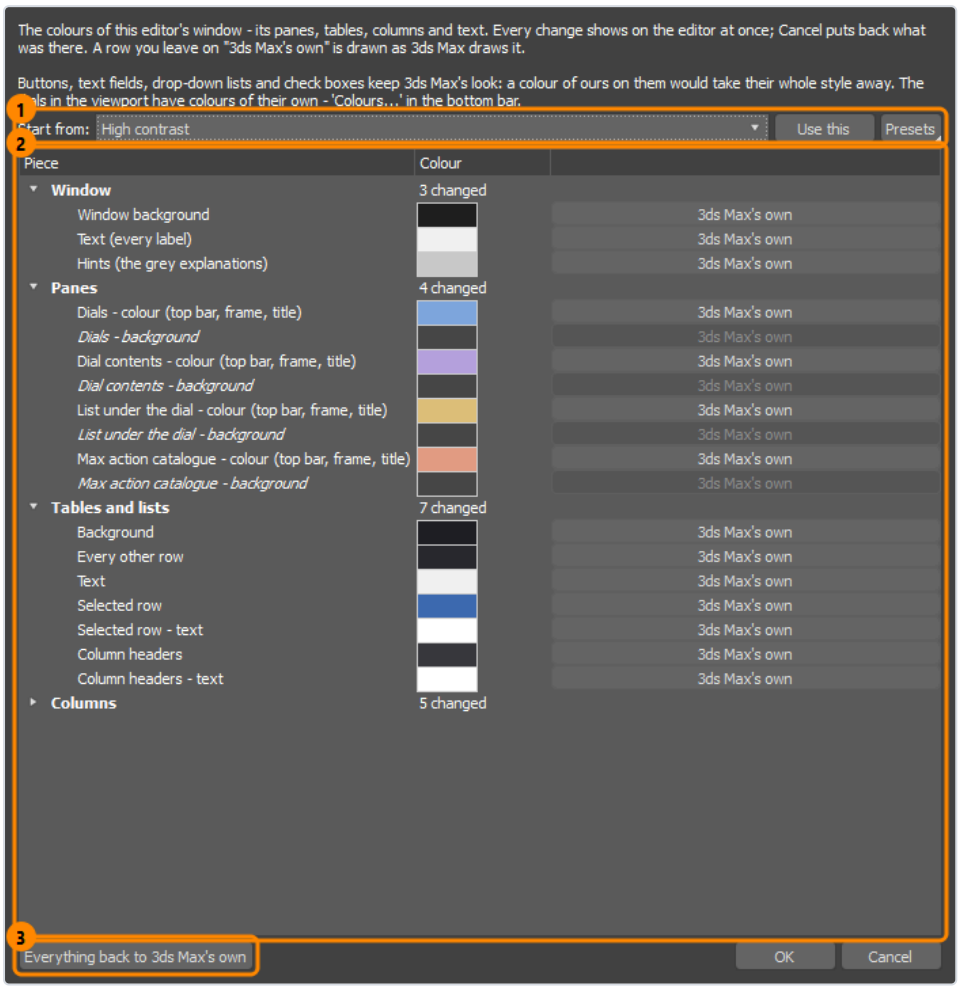
One item's caption and tile colour.

17 The editor's own colours

2 min · You will learn: to colour the editor's window - not the dials

1. Press **Editor colours...** in the bottom bar.

2. Pick a starting point in **Start from:** and press **Use this** - 21 ready-made looks (High contrast, Calm grey, Nord, Espresso...) besides *3ds Max's own*.
3. Change any piece: the window, each pane, the tables and lists, every column on its own. The editor changes as you pick. **3ds Max's own** on a row drops that colour.
4. OK keeps it - no Save needed. Cancel puts back what was there.



Editor colours...: the starting points (1), every piece of the window (2), everything back to 3ds Max's look (3).

> Buttons, text fields, lists and tick boxes keep 3ds Max's look. These colours belong to the editor, not to your dials; **Presets** saves, exports and imports them, as in tutorial 16.

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18 Print, share, reset - and add commands to a quad menu

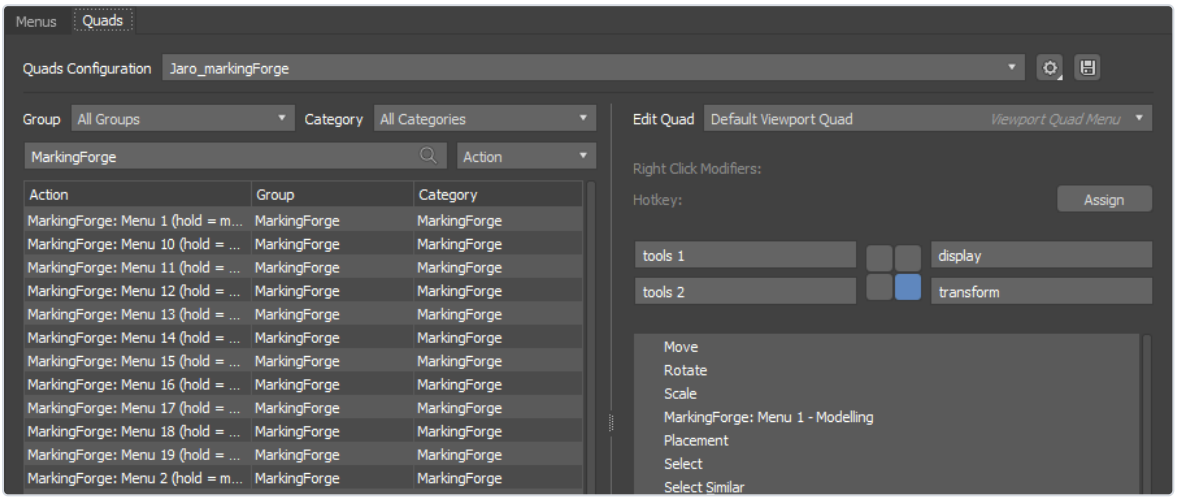
3 min · You will learn: the bottom row and 3ds Max's own menus



Cheat sheet (1), Load default dials (2), Save preset (3), Load preset (4), Save (5), Save every dial (6), Load every dial (7).

1. **Cheat sheet...** - a printable PDF of your saved dials, with keys and mouse buttons. Save first: unsaved edits are left out.
2. **Save preset...** - your layout in one file to share: the dials, their pages and their colours. **Load preset...** shows what a file holds before loading and removes scripts from other people's presets by default.
3. **Load default dials...** - the dials MarkingForge came with; by default only for the dials you have not changed. Tick *Replace...* for all of them, with their default pages and colours.

4. **Save every dial...** - each of the 24 dials and each of their pages as a file of its own, in a new folder named with the date. **Load every dial...** puts such a folder back - every dial, its pages and its colours. Scripts in the files are left out unless you choose to keep them, and nothing changes until you Save.
5. Any MarkingForge command can go into a right-click quad menu: **Customize > Menu Editor > Quads**, search `MarkingForge`, drag an action into a quad, **Done**.

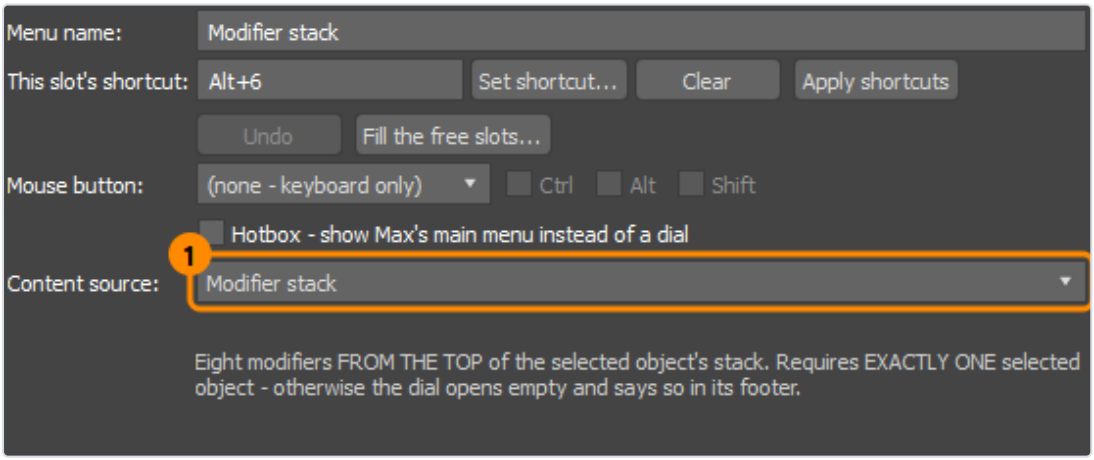


Menu Editor, Quads tab, filtered to MarkingForge.

19 Dials that fill themselves

6 min · You will learn: the modifier stack, undo, recent commands and selection sets on a dial

Content source, in *Dial contents*, decides where a dial's eight directions come from. *Items from the configuration* is an ordinary dial - the commands you put on it. The other choices **fill the dial by themselves** every time it opens - from the scene or from what you just did. The directions you had typed are kept, and come back if you switch back.



Content source, under the shortcut and the mouse button.

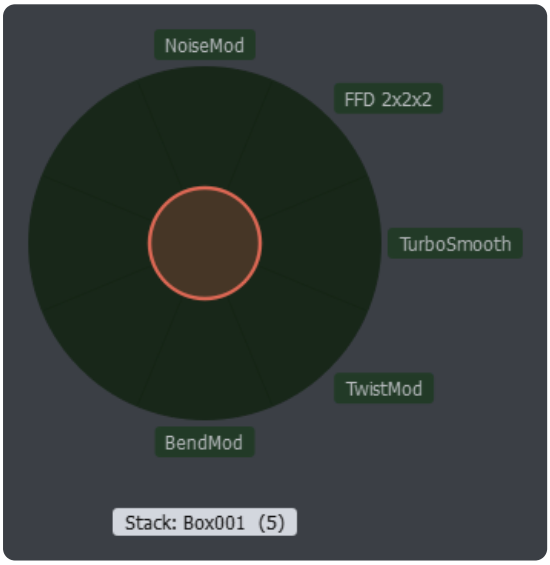
Choice	The directions are	Releasing on one	Built-in dial
Items from the configuration	what you put there	runs it	most dials
Modifier stack	the top eight modifiers of the ONE selected object	makes it the current modifier in the Modify panel	Alt+6
Undo stack	your last eight operations, by name	undoes back to it - north one step, north-east two, and so on	Alt+7

Recent commands	the eight commands you ran last, newest first (from any dial, a tap or the search)	runs it again	Alt+8
Selection sets	the scene's named selection sets	selects that set	Shift+Alt+2

Undo stack, Recent commands and Selection sets each have a switch in **Experimental features**; all three are on from the start. The **Hotbox** tick box above Content source is one more way to fill a dial: every 3ds Max menu as tiles (the last page of Alt+5).

Modifier stack - step by step

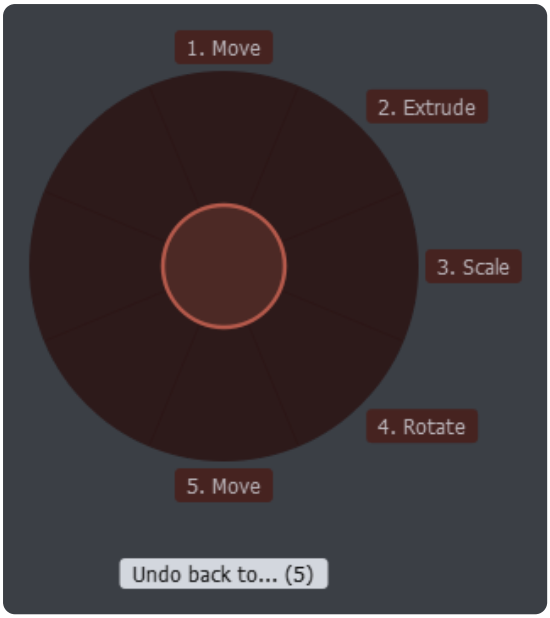
1. Pick a dial you want to turn into a stack dial (or give a free slot its own menu).
2. Set **Content source** to **Modifier stack** and press **Save**.
3. In the scene select ONE object with a few modifiers - a box with Bend, Twist and TurboSmooth.
4. Hold the dial's key: the modifiers sit on the directions, the top of the stack first. Release on Bend - the Modify panel jumps to Bend.



Alt+6 - the stack of the selected box.

Undo stack - step by step

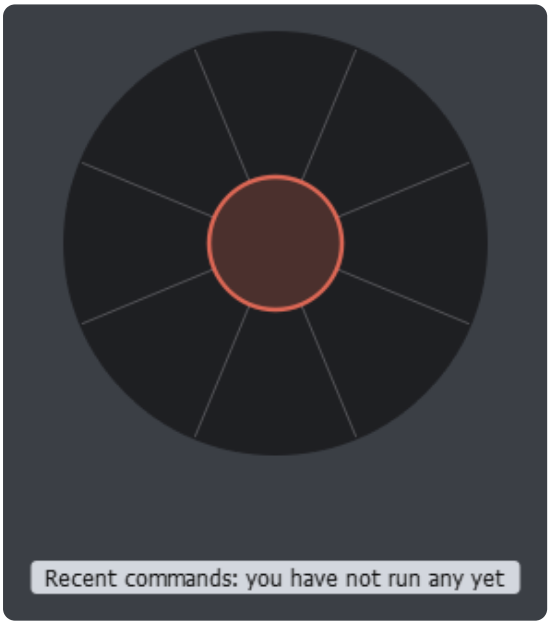
1. On a dial set **Content source** to **Undo stack** and Save.
2. Do a few things: move, extrude, add a modifier.
3. Hold the key: your operations are listed by name, the last one north. Release three directions round - three steps undone in one move.



Alt+7 - undo by name.

Recent commands - step by step

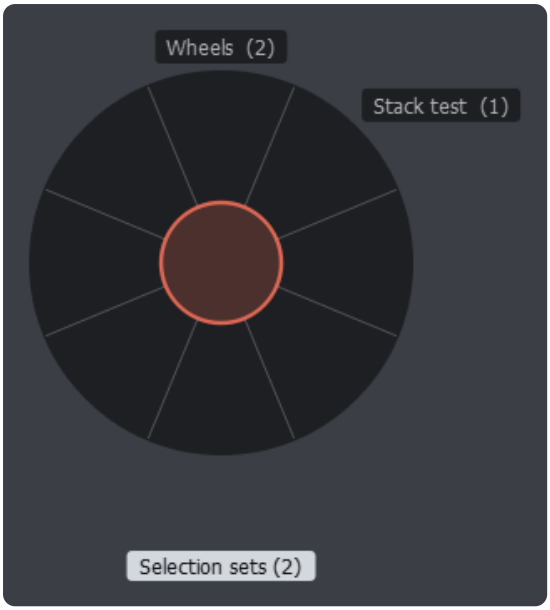
1. Set **Content source** to **Recent commands** and Save.
2. Run a few commands from any dial (a flick, a tap, a list row or the search).
3. Hold the key: they are there, newest north. A flick up repeats the last one. The history starts afresh each time 3ds Max starts.



Alt+8 - recent commands.

Selection sets - step by step

1. Select a few objects and name a selection set in the main toolbar's **Named Selection Sets** field (type a name, press Enter). Make two or three sets.
2. Set **Content source** to **Selection sets** and Save.
3. Hold the key: each set is a direction with its object count. Release on one - it is selected.



Shift+Alt+2 - the scene's selection sets.

Fill only one variant this way

1. On a dial press **Add a variant...**; in the condition window that opens set **Cursor over the window** to **Command panel** and press OK.
2. With the new variant selected, set **Content source** to **Modifier stack**.
3. Save. The same key now shows your commands over a viewport and the modifier stack over the command panel - the built-in Modelling dial does exactly this.

> A dial like this with nothing to show opens empty, and the line under it says why - *Modifier stack: select ONE object*, *Undo: nothing to undo*, *Selection sets: the scene has none*. That is not an error.

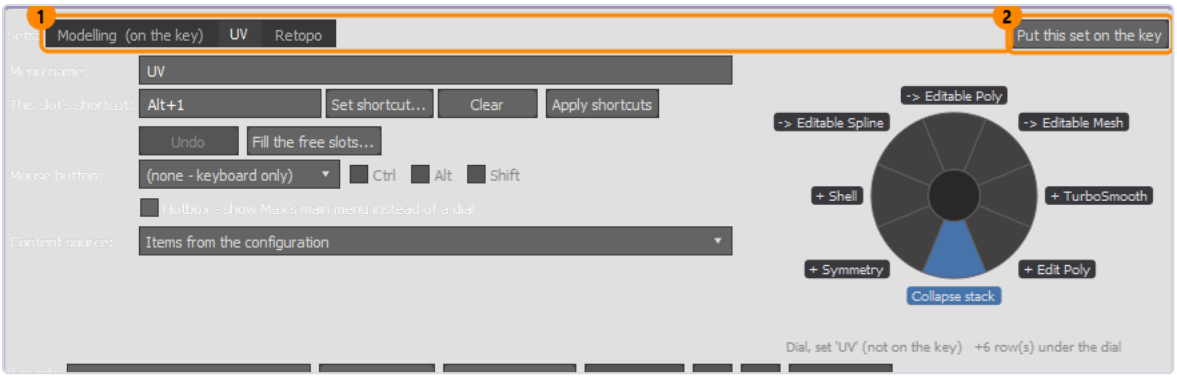
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20 Pages - several dials on one key, turned with the mouse wheel

4 min · You will learn: to keep Modelling, UV and Retopo on one key and flip between them while you hold it

A dial can hold several **sets** of contents, each with a name. They are the dial's **pages**: hold the key and turn the mouse wheel to go through them. The key, the mouse button and the other 23 dials stay as they are.

1. In the editor pick the dial - say **Menu 1** - and find **Sets of this dial** under the list of dials.
2. Press **New set...**, call it *UV* and choose **A copy of what the dial shows now** (or **An empty dial**). The new set becomes the next page, and a **tab** for it appears above the dial's contents - *Sets: Modelling (on the key) | UV*.
3. Change the new set like any dial - directions, list, variants. Click a tab to see and edit that set. Save.
4. In 3ds Max hold **Alt+1** and **turn the mouse wheel**: the dial shows its next page (wheel towards you) or the previous one, in the same place. The line under the dial says *page 2 of 2 - mouse wheel*. Release over a command to run it. Next time you press Alt+1, the dial opens on the page you used last - even after 3ds Max restarts.



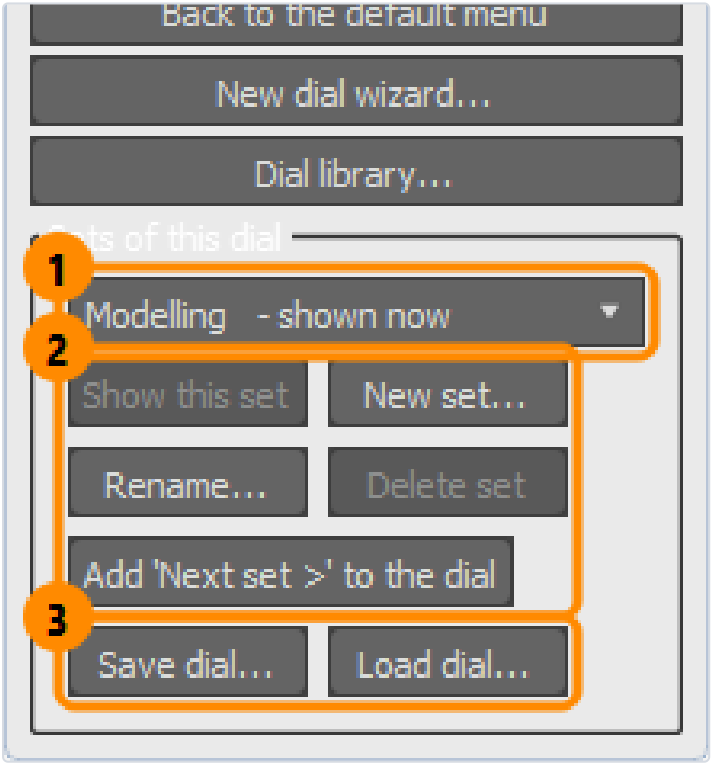
The sets of a dial as tabs (1); Put this set on the key (2).

TRY IT

Hold Alt+1, turn the wheel one notch towards you, aim at a command of the UV page and release. Press Alt+1 again: it opens on UV, where you left it. Turn back to Modelling the same way.

Which page comes first. The set marked *(on the key)* is the first page. To make another set the first one:

1. In the editor: open the set's tab and press **Put this set on the key** - or pick the set in **Sets of this dial** and press **Show this set**. Save.
2. From the dial itself: press **Add 'Next set >' to the dial**. Every set of the dial gets a *Next set* > row - pick it and the next set becomes the first page. Save.
3. With a key: **Customize > Hotkey Editor**, search `next set`, give **Menu 1 - next set** a key. There is one such action for each of the 24 dials.



Sets of this dial: the sets (1), switching and managing them (2), saving and loading one dial (3).

The wheel turns pages only while it can - while you hold the key of a dial with more than one page, on its main level, or in the hotbox when it is one of the pages (Alt+5). In a submenu, in the search or on a dial with one page, the wheel zooms and scrolls as usual. One notch turns one page.

> Switching from the dial or with a key is saved at once. If the editor is open with unsaved changes, press **Reload from file** first.

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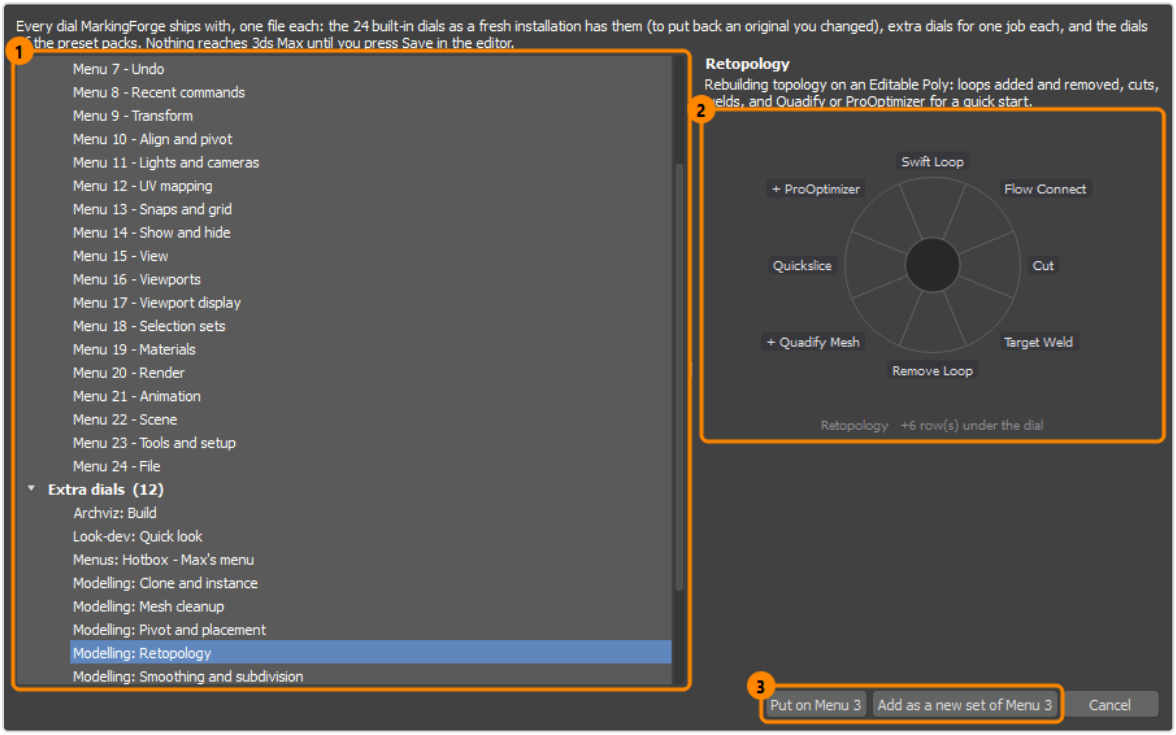
21 One dial from a file, or from the library

3 min · You will learn: to load a single dial, share one, and get an original back

Every dial can be saved as a file of its own (`.mfdial`). MarkingForge comes with 51 of them in the **dial library**: the 24 dials as they come, 12 extra dials for one job each and the 15 dials of the preset packs.

A dial from the library

1. Pick a dial in the list of dials and press **Dial library...** under it.
2. Its built-in original is selected: **Restore Menu N** puts it back as it was shipped.
3. Or choose another dial - an extra one (Mesh cleanup, Pivot and placement, Clone and instance, Smoothing and subdivision, Retopology, Quick look, Links and helpers, Build, the viewport pages of Alt+5 or the hotbox) or one from a preset pack - and press **Put on Menu N** (the key and mouse button stay), or **Add as a new set of Menu N** to keep what the dial shows and add the new dial as its next page.
4. Save.



The dial library: the dials by group (1), the one chosen (2), Put on Menu N and Add as a new set of Menu N (3).

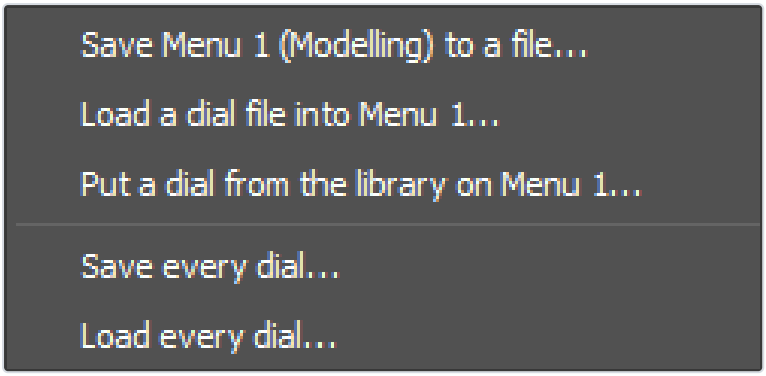
A dial from a file

1. Pick the dial and press **Load dial...** in **Sets of this dial**, and choose the `.mfdial` file.
2. If the file holds scripts, **Remove the scripts** is the default - keep them only for a file you trust.
3. Choose **A new set (keeps what the dial shows now)** or **Replace what the dial shows**. Save.

Save dial... next to it saves the set chosen in the list (the open tab) to a file, to keep it or send it.

Right-click a dial

The same is one click away: right-click the dial **in the list of dials** or **on its drawing**.



Right-click on Menu 1: save it, load a file into it, put a library dial on it - or save or load every dial.

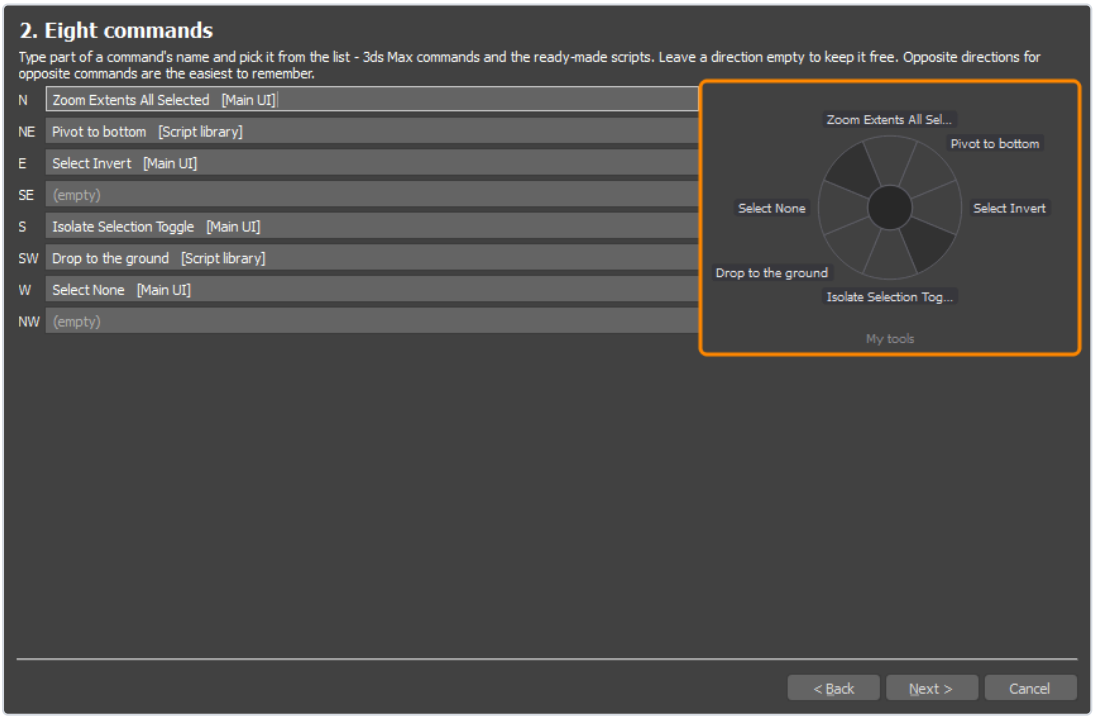
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22 A new dial in a minute - the wizard and ready-made variants

4 min · You will learn: the quickest ways to a working dial of your own

The new-dial wizard

1. Press **New dial wizard...** under the list of dials.
2. Pick the dial and a name. Leave **Add it as a new set of this dial - the key keeps what it opens now** on: what the dial shows now is kept, and your new dial is its next page.
3. For each direction type part of a command's name and pick it from the list - 3ds Max commands and the ready-made scripts are both there. The drawing beside the fields shows the dial as you fill it; a name that matches nothing turns red.
4. Finish, then Save.



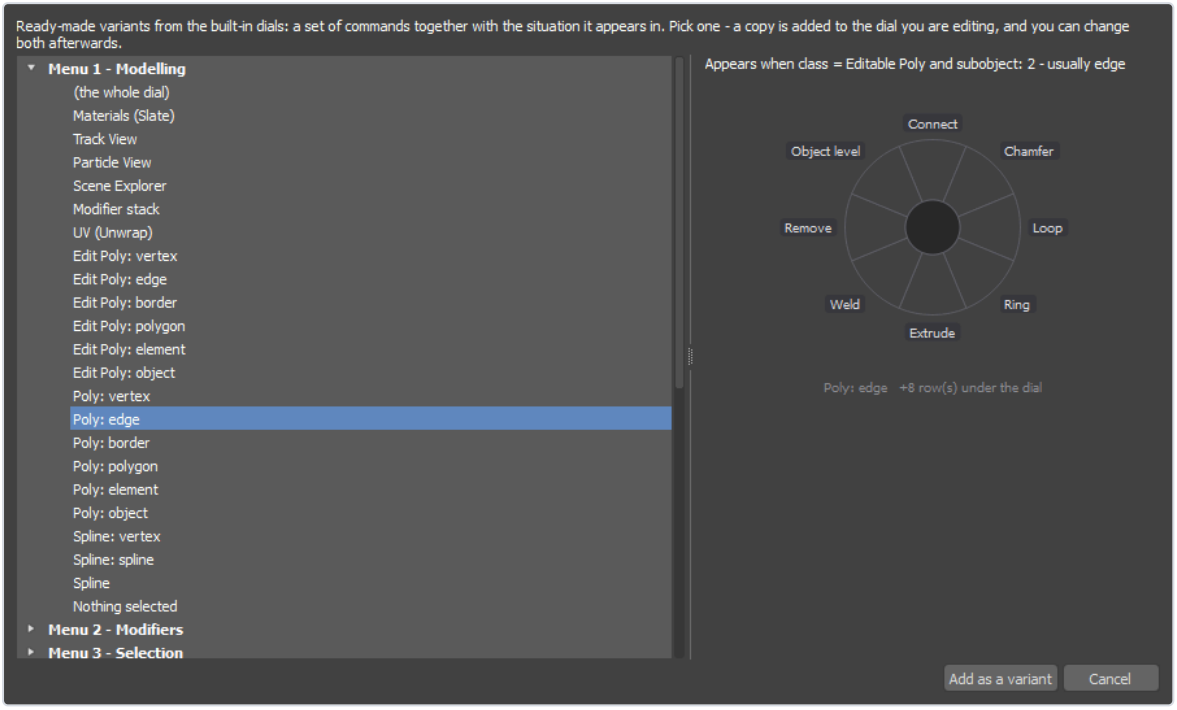
The wizard's second page: eight fields and the dial they make.

Ready-made variants

1. Pick a dial and press **From a template...** next to **Add a variant...**

2. On the left, every variant the built-in dials have - Editable Poly level by level, Edit Poly, splines, cameras, lights, the windows - and each built-in dial as a whole. On the right, when it shows and what it holds.

3. Press **Add as a variant**. It comes with its situation already set (for a whole dial you are asked for one). Change anything afterwards, then Save.



From a template: the variants of the built-in dials, with their conditions.

Learning plan and good habits

When	Do this
Day 1	Use Alt+1, Ctrl+Alt+7, Alt+3 and Alt+7 with the dial visible. Print the cheat sheet.
Days 2-5	Start flicking the five gestures of tutorial 3. Add Ctrl+Alt+1 (Transform) and Ctrl+Alt+6 (Show and hide).
Week 2	Replace directions you never use with commands you reach for in menus. Add one context variant for your most common task.
Week 3	Colour your most used dials - a ready-made look is one click. Put the rest of your frequent commands in lists; promote the ones your hand wants.
Week 4	Give the dial of your main task a second page for a job you do in bursts - UV, retopology, a dial from the library - and turn to it with the mouse wheel.

Rules for your own dials

Opposites opposite

Grow / Shrink, Hide / Unhide, Open / Save across the dial - one movement and its reverse.

Keep places stable

Do not move a command once your hand knows it. Add, do not reshuffle.

One question per dial

"What do I create", "what do I hide" - a dial of unrelated commands is hard to learn.

Specific variants first

Edit Poly at vertex level above Edit Poly, above Editable Poly.

Short labels

One or two words. The hand needs the direction, the eye only a hint.

Test with a flick

If a gesture only works while looking, the direction is in the wrong place.

If something does not work

What you see	What to do
Holding the key does nothing	**MarkingForge > Set up standard shortcuts...**, or check the key in **Customize > Hotkey Editor**.
A different dial than expected	A variant for another situation showed - the object under the cursor, the sub-object level or the window decides. Move the cursor over empty space and try again.
Alt+5 shows the viewport - where is the hotbox?	Keep holding Alt+5 and turn the mouse wheel one notch away from you.
A red caption	That command does not exist in this 3ds Max (a missing plugin). Replace it in the editor.
An empty dial	A dial that fills itself has nothing to show - select one object for the modifier stack, create named selection sets for Selection sets.
My change is not on the dial	Press **Save** in the editor.
I want the original dials back	Editor > **Load default dials...**
I changed one dial and want its original back	Pick it in the editor, **Dial library...**, **Restore Menu N**, Save.
Turning the mouse wheel over a dial zooms the viewport	The dial has one page, or you are in a submenu or the search - the wheel turns pages only on a dial with more than one page (tutorial 20).
A dial stayed on screen after you let go, and other dial keys did nothing	3ds Max sometimes misses that you let go (for example after starting a Text object). MarkingForge watches the key itself and closes the dial a moment later. If a dial still stays open, press Esc and send **MarkingForge > Diagnostics > Event log** with your report.
Letters typed over a dial vanish	That is **Type-to-search** (trick 5). Esc closes the search; switch it off in Experimental features if you do not use it.
After switching panels a dial key does nothing until you click a viewport	The keyboard got stuck on 3ds Max's menu bar (a single Alt press does that). MarkingForge gives it back by itself; if not, press Esc or click the viewport, and send **MarkingForge > Diagnostics > Event log** with your report.
Where are the other guides?	**MarkingForge > Documentation** opens the folder with all of them.

Everything else - every option, every command and more problems with their fixes - is in the **MarkingForge Reference**.

> **Still stuck?** Open an issue at github.com/Forge-Plugins/MarkingForge/issues or write to forgeplugins@gmail.com. **MarkingForge > Report a bug...** puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

Support, bug reports and feature requests

ISSUES	github.com/Forge-Plugins/MarkingForge/issues
ROADMAP	trello.com/b/7gJVqG8Y/markingforge-features-roadmap
E-MAIL	forgeplugins@gmail.com
WEB	https://forgeplugins.tech (forgeplugins.tech - more plugins by the author)

Bug reports and feature requests are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Questions, and anything you would rather not post publicly (licence, purchase), go to the e-mail address. Every message is read.

Reporting a bug

1. In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
2. Open github.com/Forge-Plugins/MarkingForge/issues and pick **Bug report** - or write an e-mail to forgeplugins@gmail.com instead. Both are links in the window that opens (since 1.5.20), one click each.
3. Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
4. Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome. What is already planned and in progress is on the **Roadmap** above - **MarkingForge > Roadmap (Trello)** opens it too.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.